

ATES-metal: Update Log

***Purpose.** This document is a running log of incremental updates to ATES-metal as the code is extended toward reproducing Huang et al. (2023) WASP-121b (see Huang_update_plan.md). Each update is a self-contained section: what changed, why, which files were touched, and the quantitative check that was run. New updates are appended as new top-level sections.*

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1. Runtime domain-extent option: Roche vs Spherical

Added 2026-06-04.

1.1 Overview

ATES historically hardcodes the full Roche potential (planetary gravity plus the stellar-tidal and centrifugal terms) and truncates the radial grid at the Hill/L1 radius. For close-in planets this Roche lobe is small; for WASP-121b the outer boundary lands at $r_{\max} \approx 2.01 R_p$. Huang et al. (2023) Case A, by contrast, is a *spherical* model (no tidal terms) extending to $\sim 19 R_p$.

To support both geometries from a single executable, a runtime option was added with two modes:

- **Roche** (default): full Roche potential, domain truncated at the Hill/L1 radius. Identical to the previous ATES behavior.
- **Spherical**: pure planetary potential $\phi = -b_0/r$ (tidal and centrifugal terms dropped), domain extended to a user-specified outer radius.

The mode is selected by two optional lines appended to `input.inp`; when they are absent the code reverts to Roche, so pre-existing input files are unaffected.

1.2 The two domain modes

1.2.1 Gravitational potential

All quantities are in code-normalized units: radius r in units of the planetary radius $R_0 = R_p$, with $b_0 = GM_p\mu/(k_B T_0 R_0)$ the surface Jeans parameter, $M_r = M_*/M_p$ the star-to-planet mass ratio, and $\tilde{a} = a_{\text{orb}}/R_0$ the orbital distance in planetary radii.

Roche mode. The dimensionless potential and its gradient are

$$\phi(r) = -\frac{b_0}{r} - \frac{b_0 M_r}{\tilde{a} - r} - \frac{b_0(1+M_r)}{2\tilde{a}^3} \left(\frac{\tilde{a} M_r}{1+M_r} - r \right)^2, \quad (1)$$

$$\frac{d\phi}{dr} = \frac{b_0}{r^2} - \frac{b_0 M_r}{(\tilde{a} - r)^2} + \frac{b_0(1+M_r)}{\tilde{a}^3} \left(\frac{\tilde{a} M_r}{1+M_r} - r \right). \quad (2)$$

The three terms are planetary gravity, stellar tidal attraction, and the centrifugal potential of the co-rotating frame.

Spherical mode. Only the planetary term is kept:

$$\phi(r) = -\frac{b_0}{r}, \quad \frac{d\phi}{dr} = \frac{b_0}{r^2}. \quad (3)$$

1.2.2 Outer boundary

- **Roche:** $r_{\max} = (3M_r)^{-1/3} \tilde{a}$, the Hill-radius approximation to the L1 distance (unchanged from before).
- **Spherical:** $r_{\max}$ is set directly by the user-supplied “Outer radius [R_p]” value (in R_p , since the grid is normalized to $R_0 = R_p$).

1.2.3 Why spherical mode must drop the tidal terms

The outer boundary cannot simply be pushed past L1 while keeping the full Roche potential. Beyond L1 the Roche potential *turns over*: the effective gravity changes sign (points toward the star), so a monotonic, steadily accelerating outward wind is no longer defined there and the sonic-point/steady-state machinery breaks down. Removing the stellar-tidal and centrifugal terms restores a monotonic potential ($\phi = -b_0/r$ increases monotonically toward zero as $r \rightarrow \infty$), which is what permits the domain to extend arbitrarily far. This is exactly the spherical geometry of Huang Case A. A guard enforces a sensible configuration: selecting Spherical mode without an “Outer radius [R_p]” > 1 stops the run with an explicit error rather than silently falling back.

1.3 Input interface and backward compatibility

Two optional lines are appended at the end of `input.inp`, after the “Force start:” line:

```
Domain mode: Spherical
Outer radius [R_p]: 19.0
```

- “Domain mode:” accepts `Spherical`; any other value (or a missing line) means `Roche`.
- “Outer radius [R_p]:” is read only in `Spherical` mode; it is ignored in `Roche` mode, where the Hill radius is computed automatically.

The reader scans the trailing lines by *keyword* (using `index(line, . . .)`), not by fixed line position, so blank trailing lines and line ordering do not matter, and `input.inp` files written before this change (which lack both lines) keep the default Roche behavior. The active mode is echoed in the setup report `ATES.out`.

1.4 Implementation

Touched files.

- `src/modules/init/parameters.f90`: new globals `spherical_domain` (logical, default `.false.`) and `r_out_user` (the outer radius in R_p).
- `src/modules/functions/grav_field.f90`: `phi` and `Dphi` branch on `spherical_domain`.

- src/modules/files_I0/input_read.f90: keyword scan of the trailing lines, and the $r_{\max}$ branch with the validity guard.
- src/modules/files_I0/write_setup_report.f90: prints the active mode to ATES.out.

Gravity branch.

```
double precision function phi(r_in)
real*8, intent(in) :: r_in
if (spherical_domain) then
  phi = -b0/r_in
else
  phi = -b0/r_in &
        -b0*Mrapp/(atilde-r_in) &
        -b0*(1.0+Mrapp)/(2.0*atilde**3.0)* &
        (atilde*Mrapp/(1.0+Mrapp)-r_in)**2.0
endif
end function phi
```

1.5 Validation on WASP-121b

Two runs were made that differ *only* in the domain-extent option; input.inp was otherwise identical (power-law SED, He/H = 0.0851, trace metals C/N/O on, “Rate/2 + Mdot/2” 2-D flux approximation). Both reached steady state via the proper criterion (momentum constant, $du < du_{\text{th}}$), not a stall or a false convergence.

mode	$r_{\max}$ [R_p]	$\log_{10} \dot{M}$	sonic [R_p]	$r(50\% \text{ HII})$	T_{peak} [K]
Roche (default)	2.01	12.68	1.825	1.178	9662
Spherical	19.00	11.81	5.270	1.152	10752
Huang Case A (ref.)	~ 19	12.57	—	—	~ 12000

$\dot{M}$ in $\log_{10} \text{ g s}^{-1}$. Convergence took $\approx 5,392$ iterations (Roche) and $\approx 154,183$ (Spherical, the larger domain).

Observations (for evaluation, not pass/fail).

- The Roche run reproduces the prior Phase-0 Case A result exactly ($\log_{10} \dot{M} = 12.68$), confirming the new optional block does not perturb default behavior.
- Spherical T_{peak} (10,752 K) is higher than Roche (9,662 K) and closer to Huang’s $\sim 12,000$ K — the extended domain lets the profile develop past the $2-R_p$ L1 clip.
- Spherical $\dot{M}$ (11.81) is lower than both Roche (12.68) and Huang (12.57), consistent with the loss of tidal enhancement, but ~ 0.76 dex below Huang.
- The sonic point moves outward from 1.83 to $5.27 R_p$ (gentler acceleration without tidal forces).

Caveats. Two confounders remain before the spherical $\dot{M}/T$ can be compared to Huang cleanly: (a) only C/N/O metals are included here, not Huang’s full Mg/Fe/Si/Ca/Na/K/S set (Phase 1); and (b) the same $N = 500$ cells now span $1 \rightarrow 19R_p$, so the inner resolution is much coarser than in the Roche run — a denser-grid spherical re-run may be needed before drawing conclusions.

1.6 Status and relation to the Huang plan

This option closes the Phase-0 geometry gap: the Roche mode is the validated baseline, and the Spherical mode provides the Case A geometry needed for later comparison. The tidal RLOF physics of Huang Phase 4 is a separate, richer treatment (substellar streamline, L1 location, terminator mapping); the present Roche mode is the existing single-streamline truncation, not the full RLOF model. The Spherical mode is the immediate tool for Case A reproduction.

2. Ionization network: add magnesium (Mg I / II / III)

Added 2026-06-05.

2.1 Overview

The first trace metal beyond C/N/O, following the Huang et al. (2023) WASP-121b species set (Mg, Fe, Si, O, C, N, S, Ca, Na, K). Per the Phase-1 plan, elements are added one at a time (order Mg $\rightarrow$ Fe $\rightarrow$ Si $\rightarrow$ Ca $\rightarrow$ Na $\rightarrow$ K $\rightarrow$ S); this section covers **Mg only**. Magnesium is solved to second ionization (Mg I / Mg II / Mg III) inside the existing coupled MINPACK ionization system, raising the species count to 18 and the ratio-unknown count to 11. The change is confined to the ionization/radiation plumbing: there are no hydrodynamic or mass-density changes, and `calc_ne/calc_rho` still count only H/He — metal electrons enter solely inside the MINPACK system.

2.2 Atomic data and rates

Sourced, not fabricated:

- **Photoionization:** Verner et al. (1996) outer-shell fits, with Mg I threshold 7.646 eV and Mg II threshold 15.035 eV (`cross_sec.f90: sigma_MgI, sigma_MgII`).
- **Recombination:** Badnell RR+DR — the same source already used for C/N/O — via `lookup_rec` cases and `rec_MgII ( $\rightarrow$  Mg I) / rec_MgIII ( $\rightarrow$  Mg II)`. This is a *deliberate deviation* from the validation-rate prescription in `Huang_update_plan.md` §1.3 (Shull & van Steenberg 1982 for Mg); it corresponds to the plan’s Phase-6 *production* rates, applied early to match the existing C/N/O treatment. The Mg ion/neutral balance below should be read as the Badnell result, not a Huang-faithful one.
- **Collisional ionization:** Voronov (1997) fits, `ion_coeff_MgI` and `ion_coeff_MgII`.
- **Line cooling:** *not yet ported* — `cool_MgI = cool_MgII = 0`. No published Mg two-level/CHIANTI cooling fit is implemented (that is Phase 2). Magnesium therefore acts only as an electron donor and a photoelectric-heating channel here, not as a coolant.

- **Charge exchange $\text{Mg} + \text{H}^+$** : omitted in this first cut.

Grid extension below 13.6 eV. Because Mg I ionizes at 7.646 eV, turning Mg on triggers the existing low-IP-metal branch (`thereis_lowIP_metal`), which extends the energy grid and the stellar SED down to 7.646 eV (`set_energy_vectors.f90`). This is required to photoionize Mg I, but has a side effect noted under Caveats.

2.3 Touched files

- `init/parameters.f90`: `n_species = 18`; `e_th_MgI/e_th_MgII`; `X_Mg`; `s_mgi/s_mgii`; `thereis_lowIP_metal`.
- `functions/cross_sec.f90`: `sigma_MgI`, `sigma_MgII`.
- `radiation/Cool_coeff.f90`: `Badnell fits`, `lookup_rec`, `rec_MgII/rec_MgIII`, `Voronov ion_coeff_MgI/MgII`, `cool_MgI/MgII (= 0, flagged)`.
- `radiation/util_ion_eq.f90`: `PH_heat_HHe` and `eval_cool` carry Mg.
- `radiation/ionization_equilibrium.f90`: Mg densities, params slots 49–55, MINPACK unknowns 10/11 (Mg II/Mg III ratios), solution unpack.
- `init/set_energy_vectors.f90`: 7.646 eV sub-grid; allocate/fill `s_mgi/s_mgii`.
- `files_IO/input_read.f90`: `X_Mg` default; `N_eq = 11`; `metals/low-IP` flags. `files_IO/metals_input_read.f90`: Mg I abundance key. `files_IO/write_output.f90`: Mg I/II/III as `Ion_species.txt` columns 17–19.
- Threading through arrays, I/O and post-processing: `init/set_IC.f90`, `files_IO/load_IC.f90`, `init/init.f90`, `ATES_main.f90`, `time_step/energy_semi_implicit.f90`, `functions/utilities.f90`, `post_process/post_process_adv.f90`.

`post_process_adv.f90` also received a latent-bug fix: its advection systems fill only the H/He equations, so they now use a local `Neq_adv` rather than the global `N_eq = 11`, which would otherwise feed `hybrd1` uninitialized metal entries.

2.4 Validation on WASP-121b

Four runs with identical `input.inp` (power-law SED, $\text{He/H} = 0.0851$, Roche domain), differing only in which metals are present. All reached steady state via $du < du_{\text{th}}$. Mg/H is conserved at the input $X_{\text{Mg}} = 3.98 \times 10^{-5}$ across the domain.

run	metals	grid floor	$\log_{10} \dot{M}$	T_{peak} [K]	$r(T_{\text{peak}})$
no metals	H/He only	13.6 eV	12.92	13722	1.703
Mg only	Mg	7.646 eV	12.92	13682	1.697
C/N/O	C, N, O	13.6 eV	12.68	9662	1.199
C/N/O + Mg	C, N, O, Mg	7.646 eV	12.76	9828	1.224
Huang Case A	full set	—	12.57	~ 12000	—

$\dot{M}$ in $\log_{10} \text{ g s}^{-1}$; r in R_p .

Observations (for evaluation, not pass/fail).

- **Clean Mg isolation** (Mg only vs no metals): $\Delta \log_{10} \dot{M} = +0.000$ dex, $\Delta T_{\text{peak}} = -40$ K. Adding magnesium alone leaves the wind essentially unchanged, as expected for a 4×10^{-5} trace species with no line cooling: it neither heats nor cools appreciably, and the soft-UV band it adds is optically thin without other absorbers.
- **Mg ionization structure**: predominantly Mg II (Mg^+) through the lower thermosphere, predominantly Mg III (Mg^{2+}) in the upper thermosphere (Mg III fraction ~ 0.47 at $1.4 R_p$ rising to ~ 0.81 at $1.9 R_p$); neutral Mg I stays subdominant. Same qualitative ordering as Huang+2023 Fig. 12.
- **MINPACK convergence**: the 11-unknown system converges without NaNs across the full temperature range.
- Mg-only $\dot{M}$ (12.92) is $\sim 2.2\times$ the Huang Case A spherical value (12.57), in the same range as the prior C/N/O Roche run; the metal-set and geometry confounds from Phase 0 still apply.

Caveats and gaps.

- **C I grid-extension confound** (C/N/O + Mg vs C/N/O). The $\dot{M}/T$ offset between these two (+0.080 dex, +165 K) is *not* a magnesium effect: extending the grid to 7.646 eV exposes **C I** (threshold 11.26 eV) to the new 11.26–13.6 eV photons the 13.6 eV-floor C/N/O run never sees, adding C I photoionization heating. This is why the clean Mg measurement uses Mg-only vs no-metals, where C is absent.
- **No Mg line cooling** ($\text{cool_MgI} = \text{cool_MgII} = 0$). Huang’s headline Mg result — Mg II h&k emission dominating the cooling at $1.15\text{--}1.4 R_p$ — cannot appear here by construction. That is Phase 2.
- **No Mg + H^+ charge exchange** yet.
- **Recombination is Badnell**, not the §1.3 validation rate.

2.5 Status and relation to the Huang plan

This completes the **Mg** step of Phase 1 (network expansion). The ionization structure satisfies the qualitative Fig. 12 ordering and the “ $\dot{M}$ within a factor ~ 2 of Case A” part of the gate; the thermal-balance part of Phase 1 is deferred to Phase 2 (Mg cooling). Next element in the planned order: **Fe**. Scratch runs and the comparison notebook live in `mg_validation/`.

3. Phase 1a: interface / data-structure refactor (no new physics)

3.1 Overview

Each metal added so far (C/N/O, then Mg in §2) widened the named-argument lists of `PH_heat_HHe`, `eval_cool`, `calc_column_dens_metals`, `write_output`, the MINPACK system `ion_system_HeH_metals`, and the driver

ionization_equilibrium.f90 by several per-ion arguments, and threaded the per-cell MINPACK coefficients through a hand-indexed flat params(60). This step converts that plumbing to a **species-metadata table + array-indexed (2D) data flow**, so that adding an element becomes “add rows to a table” instead of “thread N new arguments through M routines.” **No physics changes** — the network is still C/N/O + Mg, and the result is gated bit-for-bit against §2 (below).

3.2 What changed (Steps A–F)

- **A — Metadata module** (init/species_table.f90, new). One canonical table holds per-ion metadata (length $n_{\text{mion}} = 12$): f_sp column (mion_fsp), parent element (mion_elem), stage / charge² (mion_stage, mion_z2), photo-ionizable flag and photo-table column (mion_isphot, mion_iphot), threshold (mion_ethr), cooling flag (mion_iscool); and per-element metadata (length $n_{\text{melem}} = 4$): nuclear charge (melem_Z), neutral-ion index (melem_i0), top stage (melem_top). The canonical ion order (C I..Mg III = f_sp columns 7–18) is defined here once; element indices iel_C/O/N/Mg.
- **B — write_output 2D**. Metal ion densities pass as nm(:, 1:n_mion) and are written with an implied-do over the table ((nm(j,i)*n0, i = 1,n_mion)), not per-ion named writes.
- **C — PH_heat_HHe + column density 2D**. Optical depth, photoheating, and the photoionization rates loop over photo-ionizable ions via sigma_tab(:,mion_iphot(i)) and Nm_col, returning a 2D rate array P_m(:, 1:n_mion) (inert top stages stay zero) instead of per-ion P_CI..P_MgII arguments.
- **D — eval_cool metal channels 2D**. Recombination and collisional-ionization coefficients return as 2D arrays (rec_m, aion_m); bremsstrahlung sums mion_z2(i)*GF_elem(:,mion_elem(i))*nm(:,i) over ions. No per-ion coefficient arguments.
- **E — Generalized MINPACK system**. ion_system_HeH_metals keeps hybrdl’s fixed (N_eq,x,fvec,iflag,params) signature but (i) receives the per-cell metal coefficients through a **module-level block** in System_HeH_metals (set_metal_coeffs stores met_ntot/g0/g1/b0/b1/a1/a2), set by the driver before each hybrdl call — safe because the ionization-equilibrium cell loop is serial — and (ii) assembles fvec by **looping over elements** (two balance equations each, force-zeroing absent elements). params now carries only H/He (1–11), He/Tr (12–18), and the C/N/O charge-transfer rates (40–48); the old per-element slots 49–55 are gone. The driver builds metal densities, initial guesses, and the solution unpack by looping over n_melem/n_mion. The system size is now the formula $N_{\text{eq}} = 3 + 2*n_{\text{melem}}$ (= 11 for the current four elements), set in input_read.f90.
- **F — Generalized grid floor**. set_energy_vectors.f90 no longer hardwires the sub-13.6 eV floor to Mg’s 7.646 eV; it computes e_sub_low = min over active elements of the neutral threshold mion_ethr(melem_i0(e)) that lies below e_th_HI. The trigger flag thereis_lowIP_metal is set generically in input_read.f90 (any active element with a sub-13.6 eV neutral threshold), driven by a new canonical-order abundance array melem_ab (declared in parameters.f90, populated in input_read.f90). For the current set only C I (11.26 eV) and Mg I (7.646 eV) lie below 13.6 eV and the minimum is Mg I, so the floor is unchanged at 7.646 eV when Mg is present.

(Consequence: a future C/N/O-without-Mg run would now extend the grid to 11.26 eV, which it did not before — and once Phase 1b adds Na/K/Ca, whose neutral thresholds 5.14/4.34/6.11 eV sit below Mg's, the floor drops further.)

3.3 Gate: bit-for-bit reproduction of the C/N/O + Mg run

With C/N/O + Mg active and the same input `.inp/metals.inp` as §2, the refactored code reproduces the `mg_validation/cno_mg` reference to round-off: $\log_{10} \dot{M} = 12.76$; `Hydro_ioniz.txt`, `Hydro_ioniz_adv.txt`, and `Ion_species.txt` are byte-identical to the frozen reference. `Ion_species_adv.txt` differs at exactly one number — line 1 (innermost ghost cell), column 7 (the HeITR fraction): the reference holds 4.0048×10^{-5} left in uninitialized stack memory while the deterministic build writes 0.0; every other column of that line and every other line match. This is a pre-existing reference artifact (a HeITR ghost-cell slot never used in the H/He + metals run), not an effect of the refactor.

3.4 What is deliberately *not* generalized yet

- **Charge exchange with H** is still hardcoded for C/N/O only (the `fvec(1)` C,N,O sum and the `cxlo/cxup` mapping via `iel_C/O/N` in `System_HeH_metals`). Generalizing it is Phase 1d.
- **Per-ion atomic data and per-element abundance remain manual entries** — the `sigma_tab(:,1..8)` assignments and the `sigma_XX` cross-section functions in `set_energy_vectors.f90`, the abundance scalars `X_C/X_O/X_N/X_Mg`, the `metals.inp` keys, and the `melem_ab` population block. These are the atomic-data / abundance rows expected to grow per element in Phase 1b/1c; they are data, not interface plumbing.

3.5 Touched files

- **New:** `init/species_table.f90` (Step A).
- `init/parameters.f90`: `melem_ab(:)` abundance array; generalized `thereis_lowIP_metal` comment.
- `files_I0/input_read.f90`: `N_eq = 3 + 2*n_melem`; `melem_ab` population; generalized low-IP-metal trigger.
- `files_I0/write_output.f90`: 2D metal-column writer (B).
- `radiation/util_ion_eq.f90`: `PH_heat_HHe` + column density 2D (C); `eval_cool` metal channels 2D (D).
- `nonlinear_system_solver/System_HeH_metals.f90`: module-level per-cell metal block + element-loop residual (E).
- `radiation/ionization_equilibrium.f90`: metadata-driven density build, params packing (H/He + charge transfer only), initial guesses, solution unpack (E).
- `init/set_energy_vectors.f90`: min-over-active-metals grid floor (F).

3.6 Status and relation to the Huang plan

Phase 1a is complete: the per-element plumbing of §2 is now table-driven. Adding the Phase-1b metals should require only metadata rows plus atomic/abundance data, with **no changes to `System_HeH_metals.f90` or `ionization_equilibrium.f90`**. Next: **Phase 1b** — Si, Ca, Na, K, S added together as a uniform-template batch. Gate scratch run: `/tmp/ates_gate/cno_mg/` vs frozen reference `mg_validation/cno_mg/output/`.

4. Phase 1b: correct C/N/O cross sections and add Si, Ca, Na, K, S

4.1 Overview

Two changes are bundled here, both enabled by the Phase 1a (§3) table-driven plumbing:

1. **Corrected the C/N/O photoionization cross sections** to the full Verner, Ferland, Korista & Yakovlev (1996; VF KY96) Eq. (1) form. The previous code used the simpler Verner & Yakovlev (1995) inner-shell form (no y_0/y_1 offset/asymptote parameters) for C/N/O; that form is exact only when $y_0 = y_1 = 0$, which does not hold for the C/N/O outer shells. This is the dominant *physics* change.
2. **Added the five uniform-template metals Si, Ca, Na, K, S** in one batch. With Mg (§2) the in-code element set is now C, N, O, Mg, Si, Ca, Na, K, S — the full Huang+2023 §2.4 set *except Fe* (Phase 1c). Counts: $n_{\text{species}} = 30, n_{\text{mion}} = 24, n_{\text{melem}} = 9$.

The metal MINPACK solver was also **renamed** (below).

4.2 Variable-stage solver (2- and 3-stage elements in one packing)

Huang+2023 carries Si, Ca (like Mg) to second ionization but Na, K, S only to first. The solver keeps a uniform **two-unknowns-per-element** packing ($N_{\text{eq}} = 3 + 2 \times n_{\text{melem}} = 21$ with all nine elements) and a per-element `melem_top` flag:

- `melem_top = 2` (C, O, N, Mg, Si, Ca): solve both $X^0 \leftrightarrow X^+$ and $X^+ \leftrightarrow X^{2+}$.
- `melem_top = 1` (Na, K, S): solve only $X^0 \leftrightarrow X^+$ and **pin the unused upper unknown to zero** (`fvec(ix+1) = x(ix+1)`), keeping the Jacobian non-singular. Every `ix+1` access is guarded by `melem_top` so a two-stage element never touches its neighbor's column.

Zero-abundance elements are still force-zeroed, so the same 21-unknown system solves C-only, C/N/O+Mg, or all nine without renumbering.

4.3 Atomic data and rates (sources, not fabricated)

- **Photoionization cross sections** (`functions/cross_sec.f90`): full VF KY96 Eq. (1) form (`sigma_VF KY96`) for C I/II, N I/II, O I/II, Si I/II, Ca I/II, S I; simpler Verner & Yakovlev 1995 form (`sigma_Verner96`, $y_0=y_1=0$) for Mg I/II, Na I, K I (their published fits have $y_0=y_1=0$, so the forms agree exactly). Neutral thresholds [eV]: Si I 8.152, Ca I 6.113, Na I 5.139, K I 4.341, S I 10.36.

- **Collisional ionization:** Voronov (1997) for all new ions.
- **Recombination:** Badnell RR+DR via the swappable dispatcher, as for C/N/O and Mg — the *same deliberate deviation* from the Huang_update_plan.md §1.3 validation rates already accepted for Mg (the Phase-6 production rate applied early). Read the balances as the Badnell result.
- **Line cooling:** *not* ported (`mion_iscool = .false.`), flagged To Be Checked; the five act only as electron donors and photoelectric-heating channels (Phase 2).
- **Charge exchange with H:** *not* added; still C/N/O only (Phase 1d).

4.4 Rename: System_HeHCO → System_HeH_metals

The metal solver was named after the AIOLOS heritage (AIOLOS tracked only C and O as metals ⇒ “HCO” = H, C, O). It now carries H + He + nine metals, so the C/O-specific name was misleading. Renamed to parallel the existing System_HeH/System_HeH_TR family: module System_HeHCO → System_HeH_metals; ion_system_HeHCO → ion_system_HeH_metals; set_hco_metals → set_metal_coeffs; hco_* → met_*. Pure rename, no behavior change (§3 above is updated to the new names).

4.5 Validation on WASP-121b

Three converged runs, identical planet / SED / grid (power-law SED, He/H=0.0851, Roche domain, “Rate/2 + Mdot/2”); they differ only in code version and metal set.

key	code	metals	C/N/O σ	floor [eV]	$\log_{10} \dot{M}$	T_{peak} [K]	steps
ref	old	C,N,O,Mg	Verner96	7.646	12.76	9828	5003
A	new	C,N,O,Mg	VFKY96	7.646	12.85	12197	15382
B	new	+Si,Ca,Na,K,S	VFKY96	4.341	12.85	12135	15505

$\dot{M}$ in $\log_{10}$ g/s; Huang+2023 Case A reference $\log_{10} \dot{M} = 12.57$. ref and A share the 7.646 eV grid floor (Mg sets the minimum in the C/N/O+Mg set), so ref vs A isolates the C/N/O cross-section change with no grid-floor confound; A vs B changes both the element set and the floor (K I pulls it to 4.341 eV).

Observations (for evaluation, not pass/fail):

- **C/N/O correction (ref→A):** $\Delta \log_{10} \dot{M} = +0.090$ dex; T_{peak} rises 9828 → 12197 K and the peak moves outward from 1.22 to $\sim 1.90 R_p$. The corrected VFKY96 opacity deposits more photoheating; the hotter peak now sits near Huang’s ~ 12000 K. A genuine model change, not numerical.
- **Adding Si/Ca/Na/K/S (A→B):** $\Delta \log_{10} \dot{M} = +0.000$ dex, $\Delta T_{\text{peak}} = -62$ K, despite the floor dropping to 4.341 eV — at solar abundance and with no line cooling these five are radiatively near-passive donors.
- **Mg moves indirectly.** Mg’s own cross sections are byte-identical between ref and A, yet its ionization fractions shift by up to ~ 0.1 : an indirect response to the hotter, more extended profile (changed n_e , optical depth, photoionizing flux), not a change in Mg atomic data. Mg is coupled to the global state, hence *not* an isolated control.

- **Numerics.** Zero NaN/Inf; the 21-unknown system converges over the full T range; per-element X/H is conserved to $\sim 10^{-13}$ for all nine elements including the two-stage Na/K/S — direct evidence the variable-stage packing does not leak between element columns.
- **New-metal ionization (run B).** Si, Ca mostly *doubly* ionized in the upper thermosphere; Na, K, S mostly *singly* ionized (no second stage carried) — the qualitative Huang+2023 Fig. 12 ordering.

4.6 Caveats and gaps

- **No metal line cooling** for Si/Ca/Na/K/S (Phase 2); the thermal structure will not match Huang until Fe II/Mg II/Ca II cooling is in.
- **No charge exchange with H** for the new elements or Mg (Phase 1d).
- **Recombination is Badnell, not the §1.3 validation rate** (deliberate, as Mg).
- $\dot{M}$ is $\sim 1.9 \times$ Huang Case A (12.85 vs 12.57); the metal-set (Fe still missing) and geometry confounds from Phase 0 still apply.
- **Not done here:** Huang Fig. 12 is not digitized/overlaid (only the qualitative ordering is checked); the atomic-data values are not re-derived beyond conservation and smoothness.

4.7 Regression reference for Phase 1c

The accepted run A output is frozen at `phase1b_validation/.gate_ref_cno_mg_vfky96/output/` as the post-Phase-1b C/N/O+Mg baseline. When Phase 1c adds Fe, running with `Fe off` must reproduce it; because adding Fe bumps `n_species/n_mion` (and the `Ion_species.txt` column count), the Phase-1c check is a **round-off match on the shared physical columns**, not a byte-for-byte compare (the `mg_validation/cno_mg` byte-gate applied only to the physics-neutral Phase 1a refactor, where `n_species` was unchanged).

4.8 Status and relation to the Huang plan

This completes **Phase 1b**: the network carries C/N/O/Mg/Si/Ca/Na/K/S, the C/N/O cross sections are corrected to VFKY96, and the variable-stage solver handles 1st- and 2nd-ionization elements uniformly. The gate’s “converges/conserves/Fig. 12 ordering/ $\dot{M}$ within $\sim 2 \times$ Case A” items are addressed by the numbers above; the thermal-balance part is deferred to Phase 2 (metal line cooling). Next: **Phase 1c** — iron (Fe I/II/III), which needs non-uniform cross-section (Zatsarinny+2019) and recombination data. Scratch runs and notebook: `phase1b_validation/Phase1b_validation.ipynb`.

5. Phase 1c: add iron (Fe I / II / III)

5.1 Overview

Iron is added as the third three-stage trace metal, completing the Huang+2023 §2.4 element set (C, N, O, Mg, Si, Ca, Na, K, S, **Fe**). No solver, cooling-loop, heating, output, or post-processing code changed: the variable-stage MINPACK packing and every metal loop are generic over `n_melem`, so adding iron is purely new metadata rows plus four new rate functions. Counts: `n_species = 33`, `n_mion = 27`, `n_melem = 10`, `n_mphot = 17`; the system grows to `N_eq = 3 + 2*n_melem = 23`. Iron carries `melem_top = 2` (three stages, like C/O/N/Mg/Si/Ca).

5.2 Atomic data and rates (sources, not fabricated)

Iron is the one element whose data are *not* the uniform-template choice. Each quantity was checked against its primary source before encoding; the Voronov fit in particular was cross-checked because an earlier abandoned attempt had its P/X/K columns transposed.

- **Photoionization cross sections** (`functions/cross_sec.f90`): single-shell VFKY96 form (`sigma_VFKY96`) for Fe I and Fe II, parameters from the Verner+1996 `phfit2.f` PH2 block. Neutral thresholds [eV]: Fe I **7.902** (NIST ionization potential), Fe II **16.199**. *To Be Checked / known limitation*: Huang+2023 use the multi-shell Zatsarinny+2019 Fe I cross section; the single-shell fit here under-represents inner-shell absorption at high XUV energy, so the iron photoionization rate is a lower bound at the hardest photons (flagged inline). A multi-shell upgrade is left for later.
- **Collisional ionization**: Voronov (1997) `cfit.dat`. Fe I (`dE=7.9`, `A=2.52e-7`, `P=0`, `X=0.701`, `K=0.25`), Fe II (`dE=16.2`, `A=2.21e-8`, `P=1`, `X=0.033`, `K=0.45`).
- **Recombination**: the Huang+2023 Eqs. (5)–(6) total fits (a dielectronic Arrhenius term $A T^{-1.5} e^{-T_0/T} (1 + B e^{-T_1/T})$ plus a radiative power law $C (T/10^4)^{-\eta}$) used *directly*, not the Badnell dispatcher used for the lighter metals. Deliberate: the Huang iron rates are the validation target, so the model runs on exactly those rates; the Badnell-swap for iron is deferred to Phase 6. Fe II+*e* → Fe I: `A=2.833e-8`, `T0=5.731e4`, `B=1.383e4`, `T1=120.4`; `C=1.248e-12`, `eta=0.485`. Fe III+*e* → Fe II: `A=1.094e-5`, `T0=1.490e4`, `B=36.74`, `T1=1.153e5`; `C=1.728e-12`, `eta=0.618`.
- **Line cooling**: *not* ported (`mion_iscool = .false.` for all Fe stages) — Phase 2. This matters more for iron than for the lighter metals: Fe II is the *dominant* low-altitude line coolant in Huang+2023, so a metals-on thermal profile is *expected to run hotter than Huang* until Phase 2.
- **Charge exchange with H**: *not* added; still C/N/O only (Phase 1d). Iron's $\text{Fe}^+ + \text{H}^+ \leftrightarrow \text{Fe}^{2+} + \text{H}$ reactions (Huang Table 4) come in Phase 1d.

5.3 Touched files

Metadata rows + four rate functions + abundance/IC wiring; no solver, cooling, heating, output, or post-process changes:

- `init/species_table.f90` — `n_mion=27`, `n_melem=10`, `n_mphot=17`; Fe I/II/III rows (`f_sp` cols 31–33); `iel_Fe=10`; `melem_top` appends 2.
- `functions/cross_sec.f90` — `sigma_FeI`, `sigma_FeII`.
- `init/set_energy_vectors.f90` — `sigma_tab(:,16)=sigma_FeI`, `(:,17)=sigma_FeII`.
- `radiation/Cool_coeff.f90` — Huang recombination `alpha_rec_FeI/FeII_Huang` with `rec_FeII/rec_FeIII` wrappers; Voronov `ion_coeff_FeI/FeII`; dispatcher cases for FeII/FeIII recombination and FeI/FeII ionization. No `cool_coeff_metal` case for Fe (Phase 2).
- `init/parameters.f90`, `files_IO/input_read.f90`, `files_IO/metals_input_read.f90` — `n_species=33`; `X_Fe`; `metals.inp` keys FeI/Fe; `melem_ab(iel_Fe)`; `thereis_metals` trigger.
- `init/set_IC.f90`, `files_IO/load_IC.f90` — neutral-iron IC for `f_sp` cols 31–33.

5.4 Iron abundance

Asplund et al. (2009) solar Fe/H = 3.16×10^{-5} ($\log \epsilon_{\text{Fe}} = 7.50$). Huang+2023 adopt $4 \times$ solar metallicity for WASP-121b, so the validation run uses $X_{\text{Fe}} = 1.26 \times 10^{-4}$ by number relative to H; C/N/O/Mg are held at solar to isolate the iron change.

5.5 Validation on WASP-121b

Two converged runs (identical planet / SED / grid; power-law SED, He/H=0.0851, Roche domain, “Rate/2 + Mdot/2”) plus the frozen Phase-1b C/N/O+Mg baseline:

key	metals	iron	$\log_{10} \dot{M}$	T_{peak} [K]	$r(T_{\text{peak}})$	steps
ref	C,N,O,Mg	frozen 1b run A	12.85	12197	1.899	15382
off	C,N,O,Mg	off ($X_{\text{Fe}}=0$)	12.85	12197	1.899	15382
on	+Fe	on, $4 \times$ solar	12.87	12223	1.915	16566

$\dot{M}$ in $\log_{10}$ g/s; Huang+2023 Case A reference $\log_{10} \dot{M} = 12.57$.

Observations (for evaluation, not pass/fail):

- **Regression, force-zeroed iron (off vs ref).** The Phase-1c binary carries three extra species and two extra MINPACK unknowns, yet run off reproduces the frozen baseline on every shared column to the round-off floor: worst relative difference 2.0×10^{-13} across the seven hydro columns and 3.7×10^{-13} (NI) across the 30 shared species. The three new Fe columns are *identically zero, no NaNs*. With $X_{\text{Fe}}=0$ the iron block reduces to identity rows in the residual (`fvec=x`, rooted at zero) and contributes exactly zero electrons, so the 23-unknown system returns the 21-unknown answer — adding iron is non-perturbative when off.

- **Iron ionization structure (on).** Fe II dominates the base (fraction 0.89 at $r = 1$, peaking ~ 0.96 near $r = 1.02$); Fe III dominates the escaping flow (0.97 at the outer edge), with the Fe II/Fe III crossover near $r \simeq 1.25$; Fe I is minor everywhere (max ~ 0.11 at the base). This is the qualitative Huang+2023 Fig. 12 arrangement — Fe I confined to the base, Fe II at the base, Fe III through the outflow.
- **Iron as an electron donor.** At $4\times$ solar, iron supplies up to $\sim 26\%$ of the free-electron density, so it feeds back on the whole coupled solution, not just its own columns.
- **Switching iron on (off $\rightarrow$ on).** $\Delta \log_{10} \dot{M} = +0.02$ dex, $\Delta T_{\text{peak}} = +26$ K ($12197 \rightarrow 12223$). Small and in the *heating* direction: with no Fe line cooling yet, iron acts only as an added electron donor / photoelectric-heating channel, so the thermosphere runs slightly hotter, not cooler.
- **Numerics.** Run on converged with zero NaNs; per-element X/H is conserved to 8.3×10^{-13} across all five active elements including iron.

5.6 Caveats and gaps

- **No iron line cooling** (`mion_iscool=.false.`) — Phase 2. Because Fe II is the dominant low-altitude coolant in Huang+2023, the run-on *temperature profile is not yet comparable to Huang* and is expected to sit above it. The Phase-1c physics target is the iron *ionization structure*, not $T(r)$.
- **No iron charge exchange with H** — Phase 1d.
- **Single-shell VFKY96 Fe I/II cross sections** under-represent inner-shell XUV vs Huang’s Zatsarinny+2019 Fe I (flagged in `cross_sec.f90`).
- **Recombination is the Huang fit**, not Badnell (deliberate; the Badnell-swap is Phase 6).
- $\dot{M}$ is still $\sim 2\times$ Huang Case A (12.87 vs 12.57); the missing metal cooling and the Phase-0 geometry confounds still apply.
- **Huang Fig. 12 is not digitized/overlaid** — only the qualitative Fe II-base / Fe III-outflow ordering is checked.

5.7 Regression reference for Phase 1d / Phase 2

The run-on output (C/N/O/Mg + $4\times$ -solar Fe) is frozen at `phase1c_validation/.gate_ref_cno_mg_fe/` (input, `metals.inp`, `run.log`, `output/`) as the post-Phase-1c iron-on baseline. Phase 1d (iron charge exchange) and Phase 2 (iron line cooling) both *change* the iron ionization and the thermal structure deliberately, so this is the “before” snapshot for those comparisons, not a bit-reproduction target.

5.8 Status and relation to the Huang plan

Phase 1c **implementation is complete** and the iron ionization structure is presented for validation against Huang+2023 Fig. 12 (numbers and figures in the notebook; no pass/fail asserted here — that judgement is the reader’s). The

ionization-network half of Phase 1 now carries the full Huang element set with verified iron atomic data; the thermal-balance half remains deferred (until Fe II cooling lands in Phase 2 the metals-on temperature profile will not match Huang). Next: **Phase 1d** — charge exchange with H for all metals (Huang Table 4), including $\text{Fe}^+ + \text{H}^+ \leftrightarrow \text{Fe}^{2+} + \text{H}$. Scratch runs and notebook: `phase1c_validation/Phase1c_validation.ipynb`.

6. Phase 1d: charge exchange with hydrogen (Huang Table 4)

6.1 Overview

Phase 1d adds the charge-exchange (charge-transfer, CT) reactions of Huang+2023 Table 4 as a single table-driven module. The default active set is the 23 **Group A** metal + H / H^+ reactions, which couple each metal’s ionization balance to the H^+ fraction. A runtime switch `cx_full 1` in `metals.inp` enables the rest of the table — **Group B** (He + H, 2 rows), **Group C** (metal + He, 6 rows), **Group D** (metal + metal, 32 rows). This *replaces* the previous hard-coded Kingdon & Ferland (1996) C/N/O + H charge-exchange terms in the solver with a generic implementation covering all ten metals plus He.

Re-sourcing C/N/O to Huang Table 4 has one deliberate consequence: Table 4 tabulates only the neutral $\leftrightarrow$ singly-ionized CT for C, N, O (rows A13–A18), so the switch *drops* the doubly-ionized $\text{C}^{2+}/\text{N}^{2+}/\text{O}^{2+} + \text{H}$ recombination terms the old KF96 implementation carried. This is an intentional behavior change to match the paper; it raises the C/N/O double-ionization fractions and reshapes the thermal profile (below). No solver structure, cooling loop, heating, output, or post-processing code changed beyond the CT wiring.

6.2 Atomic data and rates (sources, not fabricated)

All 63 rate rows are transcribed verbatim from `docs/charge_exchange_table4.md`, the image-verified transcription of Huang+2023 Table 4 (23 Group A + 2 Group B + 6 Group C + 32 Group D). The paper text says “65 reactions”; the 2-row discrepancy is immaterial and noted in the reference doc.

- **Functional forms** (radiation/charge_exchange.f90): KF96 bracket $aT_4^b(1 + ce^{dT_4})e^{-E_k/T_4}$ (kf); power law $a(T/300)^be^{-E_k/T_4}$ (gp); 4th-order $\ln T$ polynomial (p4) for the radiative-CT rows of N / S / Na / K (Lin/Zhao/-Dutta/Watanabe); and the Satta (2013) $f_{\text{Si}}(T)$ and $D_{\text{CSi}}(T)$ helpers for the Si + He and C + Si rows.
- **Spot checks at $T = 10^4$ K** (against the table): $\text{Fe} + \text{H}^+ = 4.000 \times 10^{-9}$ (RV72), $\text{Fe}^+ + \text{H} = 2.011 \times 10^{-12}$, $\text{Fe}^{2+} + \text{H} = 1.260 \times 10^{-9}$ (ND87), $\text{O} + \text{H}^+ = 2.091 \times 10^{-9}$ (near-resonant, S99) — all match.
- **Rate clamp** $[0, 10^{-6}] \text{ cm}^3 \text{ s}^{-1}$: the upper bound guards the $\ln T$ polynomials, which diverge as $T \rightarrow 1$ K (never reached); the lower bound zeroes a few KF96 bracket fits that dip slightly negative below their validity floor (e.g. $\text{Mg}^+ + \text{H}^+$ at $T \lesssim 1500$ K), where the true rate is negligible.
- **Reference-doc notes**: Ca has *no* direct H charge exchange (only Group D); K has only the forward $\text{K} + \text{H}^+$ row. Reverse rates for pairs lacking direct data are Huang’s detailed-balance estimates, carried verbatim.

6.3 Touched files

A new module plus minimal wiring; the solver's old per-element C/N/O CT block is deleted, not extended:

- `radiation/charge_exchange.f90` (**new**) — the full Table-4 descriptor arrays, the `cx_rate` dispatcher, and the generic residual assembly. Entry points: `cx_init` (build the active list from `cx_full`), `cx_set_cell(T)` (per-cell rates), `cx_add_to_fvec` (add CT terms to the MINPACK residual via the `cx_fvidx` element/stage $\rightarrow$ row map), and the `cx_full` switch. The zero-abundance guard is automatic: each rate is $k n_{\text{donor}} n_{\text{acceptor}}$.
- `nonlinear_system_solver/System_HeH_metals.f90` — the hard-coded C/N/O + H charge-exchange terms are *removed*; a single call `cx_add_to_fvec(...)` now feeds CT contributions into the H, He, and every metal-stage residual row.
- `radiation/ionization_equilibrium.f90` — call `cx_set_cell(T_K(j))` before each cell's `hybrd1` solve.
- `files_IO/metals_input_read.f90` — parses `cx_full <0|1>` from `metals.inp`.
- `files_IO/input_read.f90` — if `(thereis_metals)` call `cx_init` after the input read, before the ionization sweep.

6.4 Validation on WASP-121b

Two new converged runs (identical planet / SED / grid as Phase 1c) compared against the frozen Phase-1c iron-on output as the regression reference:

key	charge exchange	$\log_{10} \dot{M}$	T_{peak} [K]	$r(T_{\text{peak}})$	steps
ref	KF96 C/N/O + H (frozen 1c on)	12.87	12223	1.915	16566
cx	Huang Table 4 Group A (default)	12.89	12955	1.778	13942
full	Huang Table 4 full, <code>cx_full 1</code>	12.88	12951	1.765	11661

$\dot{M}$ in $\log_{10}$ g/s; Huang+2023 Case A reference $\log_{10} \dot{M} = 12.57$. All three runs converged with **zero NaNs**.

Observations (for evaluation, not pass/fail):

- **H⁺ sink magnitude.** In the ionized bulk ($x_{\text{HII}} > 0.5$) the net metal charge exchange is at most 3.7×10^{-3} of radiative recombination (median 1.0×10^{-3}) — negligible as an H⁺ sink, consistent with Huang §4.2 (“Fe–H CX lower than H photoionization and recombination”). At the near-neutral base ($x_{\text{HII}} \sim 7.8 \times 10^{-8}$) the *ratio* is large (~ 118) only because H⁺ is itself a trace species there.
- **Near-resonant balance.** $\text{O} + \text{H}^+ \leftrightarrow \text{O}^+ + \text{H}$ is near-resonant and near-equilibrium at the base (gross destroy 3.66×10^5 vs produce 3.69×10^5 , net $+3.0 \times 10^3$). The net base H⁺ sink is instead driven by the fast, one-sided $\text{Fe} + \text{H}^+ \rightarrow \text{Fe}^+ + \text{H}$ channel (net -2.9×10^4).
- **C/N/O double ionization rises** (the dropped 2+ CT). Column-max fractions ref $\rightarrow$ cx: O III 0.871 $\rightarrow$ 0.946, N III 0.897 $\rightarrow$ 0.944, C III 0.851 $\rightarrow$ 0.856.

- **Iron base shift.** Fe II base fraction $0.889 \rightarrow 0.896$; Fe III essentially vanishes at the base ($0.002 \rightarrow 0.000$): $\text{Fe} + \text{H}^+ \rightarrow \text{Fe}^+ + \text{H}$ pushes iron toward Fe^+ .
- **Thermal structure.** T_{peak} rises $12223 \rightarrow 12955$ K and moves inward ($1.915 \rightarrow 1.778 R_p$). This is the *combined* effect of (i) the C/N/O re-sourcing that drops the 2+ recombination terms and (ii) the new metal + H CT. Charge exchange adds *no* radiative-loss channel, so it cannot cool the gas; Fe II / metal line cooling remains deferred to Phase 2, and the metals-on profile therefore still runs *hotter* than Huang.
- **Group A vs full Table 4 (cx vs full).** Turning on Groups B/C/D changes the bulk temperature by at most 1.5% (worst relative difference), with $\max |\Delta T| = 556$ K at $r = 1.008$; at $r = 1.02$ the full run is 357 K cooler with more base H^+ and less He II. $\dot{M}$ moves -0.01 dex and the outflow is nearly unchanged.
- **Dynamics.** v_{out} and sonic radius: ref 18.83 km/s @ $1.884 R_p$; cx 20.28 km/s @ $1.826 R_p$; full 20.40 km/s @ $1.826 R_p$. The hotter, more inward thermal peak drives a slightly faster wind with the sonic point $\sim 0.06 R_p$ further in.

6.5 Caveats and gaps

- **No metal line cooling yet** (Phase 2). Charge exchange adds no loss term, so $T(r)$ is not comparable to Huang; the C/N/O re-sourcing makes the base hotter, not cooler.
- **C/N/O 2+ + H terms intentionally dropped** to match Table 4 — a real change from the Phase-1c (KF96) reference, and the dominant driver of the O III/N III rise rather than the new metal CT alone.
- $\dot{M}$ still $\sim 2 \times$ Huang Case A (12.88–12.89 vs 12.57); the missing metal cooling and the Phase-0 geometry confounds still apply.
- **Paper lists 65 reactions, 63 transcribed** — immaterial, documented in `charge_exchange_table4.md`.
- **No Huang figure digitized/overlaid** for the H^+ budget; only the relative-magnitude statement (net CT $\ll$ recombination in the bulk) is checked.

6.6 Regression reference for Phase 2

The default Group-A run is frozen at `phase1d_validation/.gate_ref_cno_mg_fe_cx/` (input, `metals.inp`, `run.log`, output/) as the post-Phase-1d baseline. Because the C/N/O re-sourcing deliberately shifted both the ionization and the thermal structure away from the Phase-1c frozen reference, *this* snapshot — not the Phase-1c one — is the “before” reference for Phase 2 (metal line cooling, including Fe II).

6.7 Status and relation to the Huang plan

Phase 1d **implementation is complete**: the 23 Group-A metal + H reactions are active by default and the full Table 4 sits behind `cx_full`. With this, the **ionization-network half of Phase 1 is physically complete** relative to Huang+2023 §2.4 + Table 4 — the full element set, verified atomic data, and the metal–H charge-exchange coupling are all in

place. The thermal-balance half remains deferred: until metal line cooling (Fe II in particular) lands in **Phase 2**, the metals-on temperature profile stays above Huang. Next: **Phase 2** — metal line cooling. Scratch runs and notebook: `phase1d_validation/Phase1d_validation.ipynb`.

7. Phase 2: metal line cooling — validation gate closed

Phase 2 brings the CHIANTI-based metal line cooling (Mg I/II, Ca II, Na I, Fe II, on top of the C/N/O coolants) into the coupled energy balance and **closes the Phase 2 validation gate** against Huang et al. (2023) Fig. 10. The per-coolant atomic data and the CHIANTI effective-cooling tables are documented in `Update_ATES_early_phase` (Parts II–III); this section records the integrated WASP-121b gate and the diagnostic used to close it.

7.1 Active line coolants

The line-cooling ions are flagged by `mion_iscool` in `species_table.f90` (canonical order): C I, C II, O I, O II, N I, N II, Mg I, Mg II, Ca II, Na I, Fe II. Each contributes $\beta_{\text{esc}} n_e n_{\text{ion}} \Lambda_{\text{ion}}(T)$ with the effective cooling coefficient Λ_{ion} interpolated from the hardcoded `cool_logL_*` tables in `Cool_coeff.f90` ($N_{\text{COOLT}} = 41$, $\log_{10} T = 3.0\text{--}5.0$, $\Delta \log_{10} T = 0.05$). In the optically thin limit $\beta_{\text{esc}} = 1$ (see `Update_ATES_early_phase`, Part II, “Adopted assumption: 100% escape”). N I/II carry the `iscool` flag but have no tabulated line cooling ($\Lambda = 0$), so they contribute exactly zero.

7.2 Per-channel cooling diagnostic (this phase)

To close the gate quantitatively, `eval_cool` (`util_ion_eq.f90`) gained an optional `cool_chan` output — an **exact** decomposition of the total cooling rate into H/He recombination, collisional ionization, collisional excitation, bremsstrahlung, and one column per metal ion. It is read straight from the arrays `eval_cool` already builds, so the channel sum reproduces the `cool` total to machine precision (no offline re-derivation, no divergence risk). A companion writer `write_cool_breakdown_eq` (same module, called once from `ATES_main.f90` after the final equilibrium write) dumps output/`Cooling_breakdown.txt`: columns r/R_p , T , n_e , `cool_total`, the four H/He channels, then 27 metal-ion columns. The run prints `max |sum(channels)/cool - 1|` as an internal consistency check.

7.3 Validation on WASP-121b (the gate)

Converged equilibrium run (full metals, the curated `metals.inp`; same SED/grid as Phase 1d):

- Internal consistency: $\max |\sum \text{channels}/\text{cool} - 1| = 3.7 \times 10^{-16}$ (machine eps) — the breakdown is a bit-exact decomposition of the `Hydro_ioniz.txt` `cool` column.
- In the $1.15 < r/R_p < 1.4$ band the dominant radiative coolants are **Fe II** (~30%) and **Mg II** (~25%), jointly ~55%; metals are **57–85%** of all radiative cooling through the band. Fe II leads the inner edge (top coolant at $1.1\text{--}1.2 R_p$, 53% at $1.1 R_p$), Mg II takes over by $1.3\text{--}1.4 R_p$, and H recombination only overtakes above $\sim 1.5 R_p$ — the Fe II $\rightarrow$ Mg II handoff Huang report in their Fig. 10.
- $\dot{M}$ unchanged ($\log_{10} \dot{M} = 12.51$ g/s); zero NaNs.

This matches the Phase 2 gate (“a metal-driven cooling feature appears at $1.15\text{--}1.4 R_p$, Fig. 10”) and Huang_update_expected_results (Mg II + Fe II become the most important coolant in $1.15 < r/R_p < 1.4$).

7.4 Touched files

- `src/modules/radiation/util_ion_eq.f90` — optional `cool_chan` out-arg on `eval_cool` (exact per-channel split); new `write_cool_breakdown_eq` writer; `mion_fsp` added to the `species_table` use-list.
- `ATES_main.f90` — use `utils_ion_eq`, only: `write_cool_breakdown_eq`; call it after the final equilibrium `write_output`.
- Cooling coefficients themselves live in `Cool_coeff.f90` (provenance in `Update_ATES_early_phase`, Parts II–III); unchanged here.

7.5 Caveats and deferred

- The per-ion metal decomposition is exact in the default branch (`use_2lev_cool = .false.`, the WASP-121b setting). In the optional two-level fine-structure branch the per-ion terms are the resonance-line approximation and need not sum to the two-level `cool_M`.
- **Fe II** uses Boltzmann/coronal lower-level populations ($E_{\text{cut}} = 38459 \text{ cm}^{-1}$), validated vs Huang Fig. 6 coronal curve within $\sim 1.5\times$ — sufficient for the gate. The Hollenbach & McKee IR fine-structure + density-dependent two-level refinement (Huang “Sum” curve) is deferred.
- **Fe I** line cooling (NIST f -values + Van Regemorter, Huang Fig. 5) is not yet added; Fe I is a minor coolant in the gate band. (*Now added in §9; confirmed negligible — peak 3×10^{-5} of total cooling.*)
- The **CHIANTI Ly α swap** is intentionally skipped in Phase 2; the $0.35\times$ Ly α is a Phase-3 radiative-transfer suppression, not a cooling-rate change.
- Carrying metal cooling into the post-processed `_adv` profile is handled separately (runtime `pp_metals`, default frozen); see the post-process notes.

7.6 Status and relation to the Huang plan

Phase 2 is complete and the gate is closed (user-validated 2026-06-05). The metals-on thermal balance now reproduces the Huang Fig. 10 Fe II/Mg II cooling feature, closing the thermal-balance half of the network that Phase 1 left open. Next: **Phase 3** — in-code excited $\text{H}(n=2)$ + Ly α radiative transfer.

8. Phase 3a: in-code excited $\text{H}(n=2)$ + Balmer proton source and heating

Added 2026-06-05.

Phase 3a is the first half of Phase 3: an **in-code $\text{H}(n=2)$ population model** whose output feeds back into both the hydrogen ionization balance (an extra proton source) and the energy equation (Balmer photoelectric heating), fully

coupled. The second half (a Ly α radiative-transfer solve to supply a self-consistent mean intensity $\bar{J}_{\text{Ly}\alpha}(r)$) is deferred; this phase uses a **parameterized** $\bar{J}_{\text{Ly}\alpha}$ with a switch to read an external RT profile instead. Everything is **default-off**: with no stellar parameters in `input.inp` the build is byte-identical to the Phase-2 result (`use_excited_H = .false.`).

8.1 Physics and atomic data (sources, not fabricated)

- **H(2s)/H(2p) populations**: a 2×2 statistical-equilibrium system following Christie, Arras & Li (2013) — recombination cascade into $n=2$, collisional $2s \leftrightarrow 2p$ mixing, two-photon $2s \rightarrow 1s$ decay, Ly α $2p \rightarrow 1s$ with a $2p$ radiative-pumping term, and $2s/2p$ photoionization. Solved per cell (`n2_populations`).
- **Ly α mean intensity** (the $2p$ pump): parameterized as $\bar{J}_{\text{Ly}\alpha} \approx 0.1 F_{\text{LyC}} / \Delta\nu_D$ (Huang et al. 2017, Eq. 6) with a **top-down** $1/(1 + \tau_{\text{Ly}\alpha})$ **line-center attenuation** so the optically thick base ($\tau_{\text{Ly}\alpha} \sim 1.1 \times 10^8$ here) is not over-pumped. Oscillator strength $f_{\text{Ly}\alpha} = 0.4162$.
- **$n=2$ Balmer-continuum photoionization rate Γ_2** : from a **diluted stellar blackbody** above the $n=2$ threshold $E_2 = 3.40$ eV (`gamma_n2_balmer`), with $\sigma_{\text{LyC}} = 6.3 \times 10^{-18} \text{ cm}^2$ and threshold cross-section $s_2 = 1.4 \times 10^{-17} \text{ cm}^2$. Γ_2 carries the **same ξ dayside-dilution factor as ground-state EUV ionization** (`Rate/2` $\rightarrow 0.5$, `Rate/4` $\rightarrow 0.25$); for WASP-121b’s “`Rate/2 + Mdot/2`” ($\xi = 0.5$) this gives $\Gamma_2 \approx 613 \text{ s}^{-1}$, comparable to Huang’s $\approx 576 \text{ s}^{-1}$.
- **Heating**: Balmer photoelectric heating $H_{\text{pe}} = n_2 \cdot (\text{photoelectron energy} \times \Gamma_2)$ per cell (`heat_n2_balmer`); optional collisional de-excitation heating behind `incl_deexc_heat` (off by default — it overlaps the existing HI collisional-excitation cooling).

8.2 Coupling into the solver

The model runs **decoupled (lagged one outer step)** to keep it out of the per-cell Newton solve: `excited_H_update` fills two global per-cell arrays from the previous converged state, and `ioniz_eq` injects them:

- `gph_balmer_HI [s-1]` is added to the HI photoionization rate `P_HI` — an extra **proton source** ($S_{\text{proton}} = \Gamma_2 n_2$).
- `heat_balmer [erg cm-3 s-1]` is added to the photoheating rate.

`excited_H_update` is called once per outer iteration in `ATES_main.f90` before `ioniz_eq`, and once more after convergence to refresh the diagnostics. The post-loop refresh deliberately does *not* re-solve `ioniz_eq` (the Balmer heating is $\sim 10^{-4}$ of the total, not worth desyncing T/f_{sp} from the converged hydro state).

8.3 Runtime interface

Append to `input.inp` (omit for the default Phase-2-identical behavior):

Stellar Teff [K]: 6459.0

Stellar radius [R_sun]: 1.458

These set `use_excited_H` (both must be > 0). Optional further tail lines: `Jlya RT file: <path>` switches $\bar{J}_{\text{Ly}\alpha}$ to **mode 1** (read an external RT profile and use it directly, instead of the mode-0 parameterized field); `Deexc heat: True`

enables collisional de-excitation heating. As with all trailing lines the scan is by keyword, so order and blank lines do not matter.

8.4 Diagnostics

write_excited_H dumps output/Excited_H.txt (14 columns) from the converged state: r/R_p , T , n_{HI} , n_e , $\bar{J}_{\text{Ly}\alpha}$, n_{2s} , n_{2p} , $S_{\text{proton}}(n=2)$, H_{pe} , H_{dx} , $\tau_{\text{Ly}\alpha}$, then the three **proton-budget** columns added in this phase — **col12 ground-state photoionization** ($P_{\text{HI}} n_{\text{HI}}$), **col13 collisional ionization** ($C n_e n_{\text{HI}}$), **col14 recombination** ($\alpha_B n_e n_{\text{HII}}$), all volumetric [$\text{cm}^{-3} \text{s}^{-1}$]. The three rate coefficients are captured inside ioniz_eq on the converged pass (globals gph_ground_HI, cion_HI, arec_HII), so the budget is self-consistent with the solver rather than re-derived offline.

8.5 Touched files

- init/parameters.f90 — Phase-3a globals (use_excited_H, T_star_eff, R_star, gamma2_bal, hpe2_bal, jlya_mode, jlya_rt_file); feedback arrays gph_balmer_HI, heat_balmer; diagnostics Jlya_arr, n2s_arr, n2p_arr, Sproton_arr, Hpe_arr, Hdx_arr; and the proton-budget capture arrays gph_ground_HI, cion_HI, arec_HII.
- radiation/excited_hydrogen.f90 (**new**) — excited_H_update, n2_populations, gamma_n2_balmer, heat_n2_balmer, load_jlya_rt, write_excited_H.
- radiation/ionization_equilibrium.f90 — inject gph_balmer_HI/heat_balmer under use_excited_H; capture the ground-state / collisional / recombination coefficients for the proton-budget dump.
- files_IO/input_read.f90 — scan the stellar / Jlya RT file / Deexc heat tail lines; set use_excited_H, jlya_mode.
- ATES_main.f90 — in-loop excited_H_update before ioniz_eq; post-loop refresh + write_excited_H.
- Makefile — excited_hydrogen.f90 in SRC.

8.6 Validation on WASP-121b (coupling ON vs OFF)

Two converged runs, same build, differing only in the use_excited_H toggle (stellar tail lines present/absent). Both reach steady state via $du < du_{\text{th}}$ with $\log_{10} \dot{M} = 12.51$ (unchanged from Phase 2 — the coupling does not destabilize the energy solver). Diagnostics in WASP-121b/{analyze_excited_H.py, analyze_proton_budget.py, compare_on_off.py}.

Proton budget (cf. Huang Figs. 11/27), volumetric rates [$\text{cm}^{-3} \text{s}^{-1}$]:

r/R_p	S_{n2}	S_{ground}	S_{recomb}	S_{n2}/S_{ground}	S_{n2}/S_{source}
1.10	5.1e4	6.6e6	6.6e6	0.008	0.008
1.30	1.2e4	4.3e5	4.4e5	0.029	0.028
1.50	4.9e3	7.6e4	8.1e4	0.064	0.060
1.80	1.9e3	1.5e4	1.7e4	0.125	0.111
2.00	2.0e3	5.9e3	7.9e3	0.34	0.25
2.01	2.9e3	5.1e3	7.9e3	0.57	0.36

Observations (for evaluation, not pass/fail):

- **Self-consistency:** at the base $S_{\text{ground}} \approx S_{\text{recomb}}$ (6.6×10^6 vs 6.6×10^6 at $1.1 R_p$), i.e. photoionization equilibrium holds; collisional ionization is negligible (T too low).
- **n=2 share:** the n=2 Balmer channel is a minor proton source below $\sim 1.5 R_p$ ($\sim 1\text{--}6\%$ of ground-state photoionization) but climbs to **$\sim 34\text{--}57\%$ of the ground-state rate ($\sim 25\text{--}36\%$ of the total proton source) near $2 R_p$** , where the gas is already ionized and EUV is attenuated (ground-state photoionization depleted) while the thin-Ly α -fed n=2 channel is not. This is the “significant proton source approaching $2 R_p$ ” behavior Huang describe, here concentrated near $2 R_p$ rather than flat across $1\text{--}2 R_p$.

Heating budget (cf. Huang Figs. 10/26): Balmer photoelectric heating is present in the budget, peaking at **$\sim 1.8\%$ of total heating** at the Ly α photosphere ($\tau_{\text{Ly}\alpha} \sim 1$ near $r \sim 2 R_p$) and $\sim 10^{-4}$ globally — a small but nonzero entry, dominating nowhere, as expected.

ON vs OFF temperature and ionization: the coupling makes the gas slightly **more ionized** (up to $\sim 24\%$ less neutral H near $2 R_p$, Δx_{HII} up to $\sim 1.1 \times 10^{-2}$) and slightly **cooler** (up to ~ 179 K at $r \sim 2 R_p$, mean ~ 42 K). The cooling is **physical, not numerical:** the extra n=2 photoionization converts HI $\rightarrow$ HII, lowering n_{HI} ; the dominant EUV photoelectric heating scales with n_{HI} , so $\sim 24\%$ less n_{HI} at $r \sim 2 R_p$ means correspondingly less local EUV heating there, which the small Balmer H_{pe} does not offset.

8.7 Caveats and gaps

- **Parameterized $\bar{J}_{\text{Ly}\alpha}$ (mode 0).** The $0.1 F_{\text{LyC}}/\Delta\nu_D$ field with $1/(1 + \tau_{\text{Ly}\alpha})$ attenuation is a placeholder for the Phase-3b Ly α RT solve. Because our n=2 contribution is peaked near $2 R_p$ rather than flat across $1\text{--}2 R_p$, if Huang’s Figs. 11/27 are flatter/larger across the band the parameterized field is under-pumping n=2 in $1.1\text{--}1.8 R_p$; mode 1 (external RT $\bar{J}_{\text{Ly}\alpha}$) is the path to test this once a Ly α RT result exists.
- **Convergence stop is momentum-uniformity** ($du < du_{\text{th}}$), not a thermal/ionization steady-state criterion, so a small residual convergence-path sensitivity rides on top of the physical ON–OFF difference.
- **De-excitation heating off by default** (overlaps the existing HI coex cooling); left behind `incl_deexc_heat`.

8.8 Two analysis gotchas (recorded so they are not repeated)

- Hydro_ioniz.txt columns 6/7 (heat/cool) are **already physical** (write_output multiplies by q_0); do *not* multiply by q_0 again in analysis.
- Ion_species.txt column 1 is r/R_p , so $n_{\text{HI}}=\text{col2}$, $n_{\text{HII}}=\text{col3}$ — the recombination sink uses col3, not col2.

8.9 Status and relation to the Huang plan

Phase 3a is complete and the proton-source comparison is user-accepted (2026-06-05). The H(n=2) model is fully coupled into the ionization balance and the energy equation; the n=2 photoionization proton source reproduces the qualitative Huang Figs. 11/27 behavior and the Balmer photoelectric heating appears in the budget without destabilizing the solver. Next: **Phase 3b** — a Ly α radiative-transfer solve to replace the parameterized $\bar{J}_{\text{Ly}\alpha}$ with a self-consistent $\bar{J}_{\text{Ly}\alpha}(r)$ (feeding mode 1).

9. Phase 2 refinement: Fe I line cooling (NIST f -values + Van Regemorter)

Added 2026-06-05.

Fe I line cooling was one of the two coolants left deferred when the Phase 2 gate closed (§7, “Caveats and deferred”). This update adds it. CHIANTI v11 has **no Fe I model atom**, so — following Huang et al. (2023) §2.5 — Fe I cooling is built from NIST permitted-line oscillator strengths plus the Van Regemorter collision-strength approximation, with the lower levels populated in Boltzmann equilibrium over the metastable manifold. The new coolant flows through the same $\beta_{\text{esc}} n_e n_{\text{FeI}} \Lambda_{\text{FeI}}(T)$ machinery as the other Phase-2 metals.

9.1 Atomic data and method (sources, not fabricated)

The cooling coefficient per $(n_e n_{\text{FeI, total}})$ in the optically thin limit [$\text{erg cm}^3 \text{ s}^{-1}$] is

$$\Lambda(T) = \sum_{\text{lines}} f_l(T) q_{lu}(T) \Delta E_{lu},$$

$$f_l(T) = \frac{g_l e^{-E_l/kT}}{U(T)}, \quad U(T) = \sum_{\substack{\text{e3en le3els} \\ E < E_{\text{cut}}}} g e^{-E/kT}, \quad q_{lu} = 2.16 a^{-1.68} e^{-a} T^{-3/2} f_{lu},$$

with $a = \Delta E/kT$ and the absorption oscillator strength f_{lu} from the NIST A_{ki} , statistical weights, and wavelength (Huang Eq. 10).

- **Raw data:** the auditable NIST dumps fe1_nist_lines.tsv (permitted E1 lines: A_{ki} , g_i , g_k , E_i , E_k) and fe1_nist_levels.tsv (every level g , E for the partition function), fetched by fetch_fe1_nist.py.
- **Metastable manifold = EVEN-parity levels below the lowest odd level** ($z^7\text{D}^\circ$, $19351 \text{ cm}^{-1} = 2.40 \text{ eV}$). Below that energy there is no lower level for an E1 decay, so the even levels are genuinely metastable and hold the Fe I population. This is *not* a plain energy cut: the Fe II treatment can use an energy cut because its cut conveniently lands at the first odd level, but for Fe I a pure energy cut would wrongly include the radiatively-depopulated $z^7\text{D}^\circ$ odd levels. The default cut yields **17 even levels and 828 permitted cooling lines**.

- Upper levels of the cooling lines are odd (E1) and assumed to decay radiatively. The builder is `cooling_data/fe1_cooling.py`; it emits the 41-point `cool_logL_FeI(NCOOLT)` Fortran block on the ATES grid ($\log_{10} T = 3.0\text{--}5.0$, $\Delta \log T = 0.05$).

Coefficient values (per $n_e n_{\text{FeI}}$, $\text{erg cm}^3 \text{ s}^{-1}$): $\Lambda(5 \times 10^3 \text{ K}) = 1.27 \times 10^{-22}$, $\Lambda(10^4 \text{ K}) = 1.23 \times 10^{-20}$, $\Lambda(2 \times 10^4 \text{ K}) = 1.31 \times 10^{-19}$ — the rising, then flattening shape of Huang Fig. 5.

9.2 Touched files

- `cooling_data/fetch_fe1_nist.py`, `cooling_data/fe1_cooling.py` (**new**) — raw NIST fetch and the Boltzmann-manifold + Van Regemorter builder.
- `radiation/Cool_coeff.f90` — `cool_logL_FeI(NCOOLT)` table constant; new `cool_FeI(T, out)` (interpolates the table via `interp_cool_table`); dispatch cases FeI in `cool_coeff_metal` and `cool_coeff_metal_scalar`. The table-header comment block now documents Fe I alongside the other coolants.
- `init/species_table.f90` — `mion_iscool(FeI) = .true.` (canonical ion index 25), so the `eval_cool` metal-cooling loop and the per-channel breakdown both pick up Fe I automatically.

9.3 Validation on WASP-121b

Re-ran the converged full-metals WASP-121b equilibrium (same SED/grid as the Phase 2 gate) with Fe I cooling on; the per-channel `Cooling_breakdown.txt` then gives the Fe I share directly. Diagnostics: WASP-121b/`analyze_FeI_cooling.py` (left panel = the ported $\Lambda_{\text{FeI}}(T)$ coefficient read straight from `Cool_coeff.f90` vs the Fe II coronal curve, Huang Fig. 5 style; right panel = per-channel cooling vs radius, Huang Fig. 10 style), figure WASP-121b/`FeI_cooling_validation.png`. Observations (for evaluation, not pass/fail):

- **Fe I is a negligible coolant** along the whole profile: its peak share of the total cooling is 3.0×10^{-5} **at** $r = 1.06 R_p$ ($T = 8303 \text{ K}$), and it is far smaller elsewhere ($\sim 3 \times 10^{-8}$ of total at the base, where Fe II dominates). Adding Fe I does *not* perturb the gate-band Fe II $\rightarrow$ Mg II handoff.
- **Mdot unchanged** at $\log_{10} \dot{M} = 12.51$ (Fe I adds no measurable cooling), zero NaNs.
- This is the expected outcome: at WASP-121b base temperatures iron is almost entirely Fe II (Fe I fraction < 0.11 , §5), and Fe I's permitted lines have high excitation energies, so its Boltzmann-suppressed cooling cannot compete with the Fe II / Mg II forbidden + resonance lines. Fe I is carried for completeness and to mirror Huang's coolant list, not because it shapes the WASP-121b energy budget.

9.4 Status and relation to the Huang plan

Fe I line cooling is **implemented and its coefficient matches the Huang Fig. 5 shape/magnitude**; the WASP-121b run confirms it is radiatively negligible there, surfaced as a figure for the reader's judgment (no pass/fail asserted). One of the two Phase-2 deferred cooling items is now closed. The remaining one — the Fe II density-dependent (IR fine-structure) low-temperature refinement — is §10.

10. Phase 2 refinement: density-dependent Fe II line cooling (SE table)

Added 2026-06-05. (Implemented step-by-step; this section is updated as each step lands — currently: the 2-D table generator is built and verified.)

This is the second deferred Phase-2 cooling item (§7): replacing the **coronal** Fe II line-cooling coefficient with a **density-dependent** one that follows the Fe II level populations from the coronal (low- n_e) limit into the collisionally saturated (high- n_e , LTE) limit. It directly affects the cool, dense base of the WASP-121b thermosphere, where the coronal table overestimates Fe II cooling by $\sim 10^4 \times$.

10.1 Why the coronal table is wrong at the base

The Phase-2 Fe II table is the optically-thin **coronal** limit: only the ground level is populated and every collisional excitation is assumed to be followed by a radiative decay, so the cooling is strictly $\propto n_e$ (per $n_e n_{\text{FeII}}$ it is a function of T alone). That assumption fails at the cool, **dense** base of a hot Jupiter thermosphere, because the low- T Fe II cooling there is carried almost entirely by **forbidden** metastable / a⁶D fine-structure lines ($A \lesssim 5 \times 10^{-3} \text{ s}^{-1}$, far-IR 26–87 μm), whose critical densities

$$n_{\text{crit}} = A_{ul} / \sum_c n_c k_{ul}^c \sim 10^4\text{--}10^7 \text{ cm}^{-3} \quad (\text{electron collider})$$

sit far *below* the base $n_e \sim 10^8 \text{ cm}^{-3}$. Those lines are therefore collisionally **saturated** (thermalized to their LTE populations); the coronal coefficient overestimates their cooling by $\sim n_e / n_{\text{crit}}$.

10.2 The density-dependent coefficient (multilevel statistical equilibrium)

cooling_data/fe2_cooling.py solves the full CHIANTI multilevel statistical-equilibrium (SE) problem for the Fe II level populations at each (T, n_e) — 175 levels, electron-collisional excitation/de-excitation (Burgess–Tully descaled Υ from the .scups file) plus radiative decay (.wgfa A -values), normalized to $\sum_i n_i = 1$ — so the cooling smoothly interpolates between the two limits:

$$\frac{\Lambda_{3\text{ol}}}{n_{\text{FeII}}} = \sum_{u>l} n_u A_{ul} \Delta E_{ul} \quad [\text{erg s}^{-1} \text{ per ion}], \quad \Lambda_{\text{eff}}(T, n_e) = \frac{\Lambda_{3\text{ol}} / n_{\text{FeII}}}{n_e} \quad [\text{erg cm}^3 \text{ s}^{-1}].$$

At low n_e , $\Lambda_{\text{eff}} \rightarrow \Lambda_{\text{coronal}}(T)$; at high n_e it falls off $\sim 1/n_e$ as the numerator saturates. This Λ_{eff} is exactly the per- $(n_e n_{\text{FeII}})$ coefficient ATES already tabulates, so the existing assembly $\text{cool_M} = \beta_{\text{esc}} n_e \sum_i n_{\text{ion}}(i) \Lambda(i)$ carries it unchanged — the n_e in the prefactor times Λ_{eff} gives $\beta_{\text{esc}} n_{\text{FeII}} \times (\text{cooling per ion})$, i.e. the correct saturated base cooling.

Why electron-only SE is sufficient at the (mostly neutral) base. In the saturated limit the upper levels reach their **Boltzmann (LTE) populations regardless of which collider thermalizes them**, and the per-ion cooling becomes collider- and n_e -independent. At the base $n_e \sim 10^8$ already exceeds n_{crit} for the dominant forbidden lines, so the electron-only SE already reaches that LTE plateau; adding the (more abundant) neutral-H collider would only push the few still-unsaturated higher- n_{crit} lines further toward the same LTE limit — a second-order correction. An explicit H I / proton collider channel is flagged as a future refinement in the module.

10.3 Step 1 — the 2-D table generator (done)

fe2_cooling.py gained an **optimized** table builder lambda_eff_table(Tgrid, ngrid) and Fortran emitters (fortran_grid_1d, fortran_block_2d, emit_fortran_table; run python3 fe2_cooling.py --table). The optimization is required because the naive per-cell solve rebuilds a cubic spline for every one of the ~ 4300 transitions at every grid cell: instead the descaled Υ is evaluated **once per transition over the whole T-grid** (upsilon() accepts an array T), and the rate matrix is split $M = n_e Q_{\text{coll}}(T) + A_{\text{rad}}$, so the n_e -loop only reassembles and solves. The full 41×29 table builds in ~ 3 s and matches the naive per-cell solve to 1.5×10^{-15} (machine precision).

The emitted table is on the ATES temperature grid ($\text{NCOOLT} = 41$, $\log_{10} T = 3.0\text{--}5.0$, $\Delta \log T = 0.05$) $\times$ an electron-density axis cool_logne($\text{NCOOLNE} = 29$), $\log_{10} n_e = 0\text{--}14$, $\Delta \log = 0.5$ (brackets the WASP-121b run's $\log_{10} n_e \approx 7.9\text{--}9.7$ with the coronal plateau below and LTE saturation above). It is stored as cool_logL_FeII_ne(NCOOLT , NCOOLNE) = $\log_{10} \Lambda_{\text{eff}}$.

Two built-in sanity checks (printed to stderr by --table):

- **Low- n_e edge reproduces the coronal table.** At $n_e = 1 \text{ cm}^{-3}$ the SE coefficient equals cooling_effective(pop='coronal') to **0.1%** across $T = 10^3\text{--}10^5 \text{ K}$ — so swapping the 1-D coronal table for the $n_e = 1$ column of the 2-D table is a no-op, and the Phase-2 gate's coronal Fe II validation (vs Huang Fig. 6, within $\sim 1.5\times$) carries over unchanged.
- **Base suppression.** At the WASP-121b base ($T \approx 2239 \text{ K}$, $n_e = 10^8$) the coronal coefficient 2.80×10^{-20} drops to the SE value 1.79×10^{-24} — a **15,663 $\times$ suppression**. The suppression is strongest at low T (forbidden lines) and weak at high T ($T \gtrsim 3 \times 10^4 \text{ K}$, where the resonance lines with large A dominate and stay near-coronal): e.g. only $\sim 1.4\times$ at $T = 3 \times 10^4$, $n_e = 10^8$.

10.4 Step 2 — the 2-D table and bilinear interpolator in Cool_coeff.f90 (done)

The generated table and its interpolators are now in src/modules/radiation/Cool_coeff.f90:

- **Table.** integer, parameter :: NCOOLNE = 29, the cool_logne(NCOOLNE) axis ($\log_{10} n_e = 0\text{--}14$, dlog = 0.5), and the data block cool_logL_FeII_ne(NCOOLT, NCOOLNE) (a reshape([...], [NCOOLT, NCOOLNE]) with T on the inner index and n_e on the outer), pasted straight from fe2_cooling.py --table. The $n_e = 1$ column matches the existing 1-D cool_logL_FeII to the fifth decimal (first node -19.78002 vs -19.77975).
- **Bilinear interpolator.** interp_cool_table_2d(logL2d, T, ne, out) — log-log bilinear in $(\log_{10} T, \log_{10} n_e)$. The T axis reuses the uniform cool_dlogT scheme; the n_e axis uses the uniform dlogne = 0.5. Both axes clamp to the nearest edge outside the table (so n_e below the grid gives the coronal plateau, above it the LTE branch). A scalar twin interp_cool_table_2d_scalar(logL2d, Ts, nes) mirrors it bit-for-bit for the per-cell post-process solve.
- **Coolant wrapper.** cool_FeII_ne(T, ne, out) calls the 2-D interpolator on cool_logL_FeII_ne. The 1-D cool_FeII is kept as the low- n_e edge / scalar fallback; its header !To Be Checked note was rewritten to record that the density-dependent table now supersedes the coronal-only rate in the cooling assembly.

Builds clean (gfortran, full make); the new routines are additive and not yet wired, so behavior is unchanged until Step 3.

10.5 Step 3 — wiring the density-dependent Fe II into the cooling assembly (done)

The 2-D coefficient is now wired into both cooling paths so the local n_e selects the coronal $\rightarrow$ LTE-saturated branch automatically:

- **Main solver (util_ion_eq.f90, eval_cool).** n_e is already computed at the top of eval_cool (call calc_ne). Immediately after the canonical c_metal(:, i) fill loop, a short loop locates Fe II by name and overrides its column with call cool_FeII_ne(T_K, ne, ...). Nothing else changes: the existing assembly cool_M = $\beta_{\text{esc}} n_e \sum_i n_{m,i} c_{\text{metal},i}$ then multiplies by n_e , which *cancels* the $1/n_e$ in $\Lambda_{\text{eff}} = (\sum_u n_u A_{ul} \Delta E_{ul})/n_e$, leaving the correct per-Fe II-ion cooling. The saturated (LTE) populations are collider-independent, so the electron-only SE solve is exact in that limit; at low n_e , Λ_{eff} reduces to the coronal rate, so optically-thin upper-atmosphere cells are unaffected. The override flows automatically into the use_2lev_cool branch (which already reads c_metal(:, 26)) and into the per-channel cool_chan diagnostic (which reads c_metal(:, i)), so the breakdown stays an exact decomposition of cool.
- **Post-process per-cell solve (T_equation.f90).** This path computes its own local n_e (from H, He, and the frozen metal stages) and sums the metal coolants with the scalar dispatcher. The Fe II term is special-cased to call the new cool_FeII_ne_scalar(TT, ne) (a thin wrapper over interp_cool_table_2d_scalar), so the post-process temperature balances the same density-dependent Fe II cooling.
- **species_table.f90.** The !To Be Checked Fe II comment was rewritten to state that eval_cool now applies the density-dependent SE coefficient (no longer “deferred”).

Builds clean (gfortran, full make, all four touched modules relink).

10.6 Step 4 — WASP-121b run and the cooling budget (numbers for the reader)

A fresh WASP-121b run (solar metals, pp_metals 1, otherwise the same input.inp as the Fe I-cooling run) was compared against the snapshot of the previous (coronal Fe II) run kept in output_coronalFeII/. Both share the same grid, so Cooling_breakdown.txt columns are differenced directly (analyze_FeII_density_correction.py $\rightarrow$ FeII_density_correction_compare.png).

At the dense base ($r = 1.0$, $T = 2358\text{K}$ fixed by the lower BC). The Fe II line cooling drops from the coronal 2.23×10^{-4} to the SE $1.71 \times 10^{-8} \text{ erg cm}^{-3} \text{ s}^{-1}$ — a $\approx 1.3 \times 10^4 \times$ suppression, consistent with the offline fixed- (T, n_e) check in Step 1 ($15,663 \times$ at $T = 2239\text{K}$, $n_e = 10^8$). Because coronal Fe II carried **99.1%** of the base cooling, the *total* radiative cooling at the base falls $\approx 122 \times$ ($2.25 \times 10^{-4} \rightarrow 1.84 \times 10^{-6}$); Fe II’s share there is now **0.9%**. The previous run’s post-process printed “35 of 502 cells fell back to eq T (stiff base band)”; with the spurious base overcooling removed that message is **gone** — the per-cell T solve no longer stiffens at the base.

Transition region 1.15–1.4 R_p (Huang Fig. 10). The band-integrated metal-line cooling share in the new run is

coolant	share
Mg II	44.0%
O II	18.9%
Fe II	13.4%
O I	7.3%
C II	6.3%
Ca II	5.1%

so **Mg II is now unambiguously the dominant metal coolant** across the transition region, with Fe II demoted to a $\sim 13\%$ contributor — the qualitative ordering Huang’s Fig. 10 shows. (In the old coronal run Fe II still topped the budget out to $\sim 1.2 R_p$.)

Global response. Removing the base overcooling raises the heating efficiency, so the steady-state mass-loss rate rises from $\log_{10} \dot{M} = 12.51$ to **12.80 g/s** ($\approx 2\times$). The self-consistent wind is correspondingly denser and, above the base, slightly *cooler* (e.g. $T(1.05)$: $7810 \rightarrow 5744$ K) — the freed energy goes into mechanical escape and adiabatic expansion rather than being radiated, the expected energy-limited behavior. Whether this $\dot{M}$ and the transition-region budget match Huang’s WASP-121b quantitatively is left to the reader’s judgment against the paper; the figure and table above are surfaced for that comparison, not asserted as a pass.

Caveat on the figure’s panel (b). The “suppression factor” there is the *ratio of Fe II cooling between the two runs*, not a fixed-condition coefficient ratio. At the base (BC-fixed T) it equals the coefficient suppression ($\sim 1.3 \times 10^4$); above $\sim 1.2 R_p$ the coronal and SE coefficients converge (the hot resonance lines dominate and stay near-coronal), so the residual ratio there reflects the changed — denser — wind profile, not the coefficient.

11. Post-process refinement: re-solve metal ionization in the advected wind (pp_metals 2)

Added 2026-06-05.

The `_adv` profiles that `TPM.py` reads are built by `post_process_adv.f90`, which re-solves the *per-cell temperature* at the advection-corrected structure (the wind is advected, then its $T(r)$ is re-converged against heating/cooling). Until now the metal *ion* densities carried into those profiles were handled in one of two ways, selected at runtime by `pp_metals` in `metals.inp` ($\rightarrow$ `pp_metal_mode`):

- `pp_metals 0` — **metal-free legacy** `_adv` (the pre-metals behavior).
- `pp_metals 1` — **frozen equilibrium** metals (the default): the converged equilibrium metal stages are carried into `_adv` unchanged, so they track the *equilibrium* structure, not the (slightly hotter, more ionized) advected wind.

This step implements the deferred third option:

- `pp_metals 2` — **re-solve**: the metal ionization balance is recomputed in each cell at the advection-corrected H/He and the post-process temperature, so the metal stage split tracks the advected wind self-consistently.

11.1 Implementation — reuse the coupled residual with H/He pinned

The metals are pure equilibrium species (no advection equations of their own): in the main loop they are solved *coupled* with H/He through `ion_system_HeH_metals + hybrd1` (photo + collisional ionization + recombination + charge exchange, no time derivative). “Re-solve” therefore means re-evaluating that same equilibrium balance at the post-process local T and the advected H/He — *not* a new physics path.

Rather than duplicate the residual, mode 2 reuses it verbatim through a thin wrapper, `ion_system_metals_pp` (new, in `post_process_adv.f90`):

```

subroutine ion_system_metals_pp(N_in,x,fvec,iflag,params)
...
call ion_system_HeH_metals(N_in,x,fvec,iflag,params) ! full coupled residual
fvec(1) = x(1) - pp_xHII_fix      ! pin HII fraction
fvec(2) = x(2) - pp_xHeII_fix     ! pin HeII fraction
fvec(3) = x(3) - pp_xHeIII_fix    ! pin HeIII fraction
end subroutine

```

The wrapper overwrites only the three H/He rows with identity equations that pin H/He to their advection-corrected fractions (stored in module-level `pp_xHII_fix` / `pp_xHeII_fix` / `pp_xHeIII_fix`), while leaving the $2n_{\text{elem}} = 20$ metal-stage rows untouched. So `hybrd1` moves only the metal unknowns; H/He stay fixed at the advected wind. Two properties fall out for free:

- **Self-consistent n_e** . The reused residual computes the electron density *internally* from the pinned H/He plus the metals it is currently solving — exactly as the equilibrium solver does — so the re-solve sees the same n_e (metals) feedback, not a frozen n_e .
- **Charge exchange with the advected H**. The Kingdon & Ferland $\text{H} \leftrightarrow \text{metal}$ couplings inside `ion_system_HeH_metals` see the *pinned* H^+/He^+ densities, so the metal balance reacts to the advected hydrogen ionization, not the equilibrium one.

11.2 The per-cell sweep in `post_process_adv`

Mode 2 adds one sweep over cells, placed after the post-advection `calc_ne` and before the photoheating refresh, inside the same k -loop that re-converges T , so metals, H/He and T are driven to a *joint* fixed point. Per cell j (skipping $n_h(j) \leq 0$):

1. `cx_set_cell(T_K(j))` — charge-exchange rate coefficients at the local T .

2. Build the per-element coefficients (canonical order) from the existing post-process rate arrays: $\text{meg_g0} = \text{P_m}$ (photoionization), $\text{meg_b0} = \text{aion_m_pp}$ (collisional ionization), $\text{meg_a1} = \text{rec_m_pp}$ (recombination), and the second-stage g1/b1/a2 where $\text{melem_top} \geq 2$; hand them to `set_metal_coeffs`.
3. Pin $\text{pp_xHII_fix} = n_{\text{HII}}/n_h$, $\text{pp_xHeII_fix} = n_{\text{HeII}}/n_{he}$, $\text{pp_xHeIII_fix} = n_{\text{HeIII}}/n_{he}$; set `params(7) =  $n_h$` , `params(8) =  $n_{he}$` .
4. Seed `sys_x` with the pinned H/He fractions plus the current metal split and call `hybrd1(ion_system_metals_pp, . . .)`.
5. **Conserve element totals:** the pre-solve totals `nm_tot_pp(:,im)` (summed over each element's stages once, before the k -loop) are multiplied by the re-solved fractions to write back `nm_w`. Only the *stage split* changes; the element abundance is untouched.

Supporting edits: the use block pulls in `ion_system_HeH_metals`, `set_metal_coeffs`, and `cx_set_cell`; `params` was widened from dimension(25) to dimension(60) to match the residual's dummy length (only entries 1–11 are consumed by the metal solve); module-level `pp_x*_fix` and locals `nm_tot_pp / meg_* / i0`, `top`, `im` were added. Builds clean (gfortran, full make).

11.3 Clean A/B and what changes

Because the **equilibrium** solve in the main RK loop is mode-independent and deterministic, a full `pp_metals 2` run reproduces `Hydro_ioniz.txt`, `Ion_species.txt`, and `Cooling_breakdown.txt` **byte-identically** to the `pp_metals 1` run (verified with `diff -q`); *only* the `*_adv.txt` profiles differ. So the A/B isolates exactly the post-process metal treatment. The mode-1 default is preserved in `output_ppm1_dd/`, the mode-2 run in `output_ppm2_resolve/`, and the canonical `output/` is left at the mode-1 default (metals.inp reset to `pp_metals 1`); `analyze_metal_resolve.py` differences the two and writes `metal_resolve_compare.png`.

Singly-ionized fraction X^+/X_{tot} in the `_adv` wind (eq frozen $\rightarrow$ re-solve). Only the elements that move appreciably are tabulated (Mg and Na are unchanged to $\leq 2\%$); the full per-element output is in the analysis script.

r/R_p	T_{ad3} [K]	O	N	Ca	Fe
1.05	5798	.00 $\rightarrow$.00	.01 $\rightarrow$.02	.64 $\rightarrow$.59	1.00 $\rightarrow$ 1.00
1.10	7984	.02 $\rightarrow$.01	.12 $\rightarrow$.13	.65 $\rightarrow$.60	.99 $\rightarrow$.99
1.15	9044	.10 $\rightarrow$.06	.38 $\rightarrow$.40	.66 $\rightarrow$.59	.99 $\rightarrow$.99
1.20	10035	.37 $\rightarrow$.26	.51 $\rightarrow$.47	.61 $\rightarrow$.54	.93 $\rightarrow$.96
1.30	10274	.34 $\rightarrow$.34	.35 $\rightarrow$.34	.38 $\rightarrow$.38	.39 $\rightarrow$.40
1.40	10680	.23 $\rightarrow$.23	.23 $\rightarrow$.23	.25 $\rightarrow$.25	.18 $\rightarrow$.19

Largest density change. The per-element scan of $\max |\Delta n(X^+)/n(X^+)|$ over the wind 1.0–1.48 R_p :

element	max rel. change	at r/R_p	$n(X^+)$ frozen $\rightarrow$ resolve
Fe	+9.5%	1.232	$1.19 \times 10^5 \rightarrow 1.30 \times 10^5$
Ca	-11.9%	1.191	$1.24 \times 10^4 \rightarrow 1.09 \times 10^4$
Na	-6.2%	1.000	$3.50 \times 10^6 \rightarrow 3.28 \times 10^6$
C	-2.1%	1.216	$1.22 \times 10^6 \rightarrow 1.19 \times 10^6$
Mg	-2.0%	1.000	$9.82 \times 10^7 \rightarrow 9.62 \times 10^7$

The corrections are **modest and concentrated in the 1.1–1.25 R_p ionization transition**, where T_{ad3} runs a few hundred K hotter than T_{eq} : **Fe II** is *higher* across the 1.18–1.27 R_p shoulder (blue above red in the figure), **Ca II** is *lower* by ~ 5 –7 points through 1.05–1.25 R_p , and **every element converges back to the equilibrium split** in the outer wind ($\gtrsim 1.27 R_p$) where T_{ad3} rejoins T_{eq} . The post-process printed **no** stiff-base T -fallback message in mode 2 (the density-dependent Fe II cooling of the previous section already removed the base overcooling).

Tiny-denominator artifacts (not physical). The same scan reports **O** +6538% at $r = 1.000$ and **N** +17.9% at $r = 1.013$. These are at the dense base where O and N are essentially *fully neutral* (O II fraction ≈ 0.00 in the table), so $n(\text{O II})$ is a vanishingly small number; a tiny absolute shift ($n^+ \approx 9.7 \times 10^2 \rightarrow 6.4 \times 10^4 \text{ cm}^{-3}$, against an O total $\sim 10^8$) reads as a huge *relative* change but is negligible in absolute and in any observable. They are denominator artifacts, not a physical re-solve effect.

Reader’s judgment, not a pass. The re-solve makes the metal stage split in the `_adv` profiles consistent with the advected wind rather than the equilibrium structure, with the physically expected sign (more ionization where the advected wind is hotter) and magnitude (single-digit-percent, transition-localized). The default remains `pp_metals 1` (frozen): the deliberate, documented choice (the equilibrium split is the converged one, and `TPM.py`’s lines are dominated by H and He). Mode 2 is provided for studies that need the metal ionization to track the advected wind. The table and `metal_resolve_compare.png` are surfaced for that comparison, not asserted as a validation pass.

12. Phase 3b: in-code $\text{Ly}\alpha$ radiative transfer ($\bar{J}_{\text{Ly}\alpha}$ via escape probability)

Phase 3a drove the $\text{H}(n=2)$ population with a *parameterized* $\text{Ly}\alpha$ mean intensity $\bar{J}_{\text{Ly}\alpha} = 0.1 F_{\text{LyC}} / \Delta\nu_D / (1 + \tau_{\text{Ly}\alpha})$ (mode 0). That estimate peaks the $n=2$ population near $2R_p$ and *under-pumps* it across 1.1–1.8 R_p , leaving a `jlya_mode=1` hook (`load_jlya_rt`) to ingest a real RT field. Phase 3b supplies the real $\bar{J}_{\text{Ly}\alpha}(r)$. Three findings reshaped the original plan (a Monte Carlo with an offline two-pass and a β -on-cooling term):

1. **Method $\rightarrow$ escape probability, not Monte Carlo.** The WASP-121b column is extremely thick: $\tau_0(\text{Ly}\alpha) \approx 5.5 \times 10^7$ at the base (Voigt $a \approx 5 \times 10^{-4}$, so $a\tau_0 \approx 10^4 \gg 1$). A faithful acceleration-free MC needs $\sim \tau_0 \approx 10^7$ – 10^8 scatterings/photon (intractable), and core-skipping is both forbidden and incompatible with the core-dominated pumping field. For $a\tau_0 \gg 1$ the Neufeld (1990)/Harrington analytic wing-escape solution is accurate, so we use a local escape-probability source function.
2. **In-code, not offline.** Being a cheap formula, $\bar{J}_{\text{Ly}\alpha}$ is evaluated every timestep inside the run (`jlya_mode=2`, the same lagged-explicit decoupling as the Phase-3a Balmer feedback); a single run is self-consistent. The

jlya_mode=1 option to read an external (e.g. real MC) profile is kept.

3. **No β -on-cooling term.** From Huang §2.5/Fig. 7 directly: the “Ly α cooling 10 $\times$ lower” is the Black (1981) vs CHIANTI *collisional-excitation rate-coefficient* difference, not photon trapping. ATES already uses the CHIANTI-consistent Cen coex_rate_HI (§7), so there is no 10 $\times$ to apply. The trapping feedback is instead the Phase-3a H($n=2$) heating (photoelectric Hpe + collisional de-excitation Hdx), which strengthens once $\bar{J}_{\text{Ly}\alpha}$ is correct. The scaffolding β -on-coex hook was reverted.

12.1 The escape-probability field (lya_rt.f90 : jlya_escape_prob)

Per cell, from the converged T , $n_{\text{HI}} = n_{1s}$, n_{HII} , n_e :

- $\Delta\nu_D = \nu_{\text{Ly}\alpha} \sqrt{2kT/m_{\text{H}}/c}$; Voigt $a = (A_{2p1s}/4\pi)/\Delta\nu_D$; optical depth $\tau(r) = \int_r^\infty n_{1s}\sigma_0 dr'$, $\sigma_0 = C_{\text{Ly}\alpha}/\Delta\nu_D$.
- Wing escape probability $\beta = \pi^{-1/4} \sqrt{a/\tau} (\leq 1)$: escape once the photon diffuses to $x_* = (a\tau/\sqrt{\pi})^{1/2}$ where the wing optical depth is unity.
- Internal sources $P = \alpha_B n_e n_{\text{HII}} + q_{1s2p} n_e n_{\text{HI}}$ (recombination cascade + electron-impact $1s \rightarrow 2p$). The escape-probability closure $\bar{J} = S(1 - \beta)$ with the trapped pile-up $n_{2p} = P/(A_{2p1s}\beta)$ gives $J_{\text{int}} = (2h\nu^3/c^2)(g_{1s}/g_{2p})P(1 - \beta)/(A_{2p1s}\beta n_{1s})$. The $(1 - \beta)$ factor makes $J_{\text{int}} \rightarrow S$ when thick and $\rightarrow 0$ when thin ($\beta \rightarrow 1$), instead of $S \propto 1/n_{1s}$ diverging as the gas ionizes in the outer wind (which spuriously raised $\bar{J}$ outward).
- Stellar beam (trapped): Ly α is *scattered*, not destroyed, so it is not killed by $e^{-\tau}$; the broad stellar line (half-width $\Delta v_s \approx \pm 70 \text{ km s}^{-1}$) penetrates through its wings. A photon at offset x reaches r if $|x| > x_1 = \sqrt{a\tau/\sqrt{\pi}}$; for a Gaussian stellar profile of width $X_s = \Delta v_s/v_{\text{th}}$, $T_*(r) = \text{erfc}(x_1/(\sqrt{2}X_s))$. The profile-averaged mean intensity uses the *stellar* line width $\Delta\nu_* = \nu_0 \Delta v_s/c$ (constant), NOT the planetary $\Delta\nu_D$ ($\bar{J} = \int J_\nu \phi d\nu \approx J_\nu(\nu_0)$ for a stellar field flat across the narrow planetary line; dividing by $\Delta\nu_D$ makes J_{star} rise as the wind cools). A bounded buildup $E = 1 + (\text{boost} - 1)(1 - \beta)T_*$ enhances the penetrating beam in the thick wind and $\rightarrow 1$ when thin or at the deep edge, avoiding both the $1/\beta$ over-count (which blew the $n=2$ source to 6300 $\times$ ground-state) and a flat cap. Hence $J_{\text{star}} = \xi F_* T_* E / (4\pi \Delta\nu_*)$, ξ the dayside dilution. $\bar{J}_{\text{Ly}\alpha} = J_{\text{int}} + J_{\text{star}}$.

New input.inp lines (default-off $\Rightarrow$ byte-identical to 3a; a 3b-0 scaffolding step verified this, frozen reference output_pre3b_baseline/):

```
Jlya escape-prob: True           # jlya_mode = 2 (in-line escape-prob RT)
Stellar Lya flux [erg/cm2/s]: 1.0e5 # incident stellar Lya at the planet (x0.35)
Lya stellar halfwidth [km/s]: 70   # broad stellar line Dv_s (penetration + Dnu_star)
Lya stellar boost [-]: 5           # bounded trapping-buildup cap (left un-tuned)
Deexc heat: True                  # Phase-3a collisional de-excitation heating
```

12.2 WASP-121b validation setup (spherical Case A, matched to Huang)

Huang’s Fig. 11 is **Case A** (spherical, no RLOF), so the comparison must match his geometry *and* radial frame — which exposed a setup error: the run had been Roche-truncated at $\sim 2R_p$, and ATES’s base ($r=1$) sits at **1 μbar** ($n_0 = 3.1 \times 10^{12}$,

$T = 2358\text{ K}$), but Huang defines R_p at the **4 mbar transit radius** and places his $1\text{ }\mu\text{bar}$ hydro bottom at $\sim 1.25 R_p$ (Fig. 9 top-pressure axis / Fig. 12 green band; the earlier “ $1.46 R_p$ ” note was wrong). The setup was corrected to spherical Case A with R_0 set to Huang’s $1\text{ }\mu\text{bar}$ radius ($R_0 = 1.25 \times 1.766 = 2.2075 R_J$, planet mass unchanged $\Rightarrow$ correct base gravity) and outer radius $15.2 (= 19 R_p^{\text{Huang}})$; ATES r maps as $R_{\text{Huang}} = 1.25 r$. The Case-A $\dot{M} = 0.077 M_p/\text{Gyr}$ is within $\sim 1.5\times$ of Huang’s 0.052 (Table 3).

12.3 Trend vs Huang Fig. 11 (qualitative — exact match not expected)

Huang’s $\bar{J}_{\text{Ly}\alpha}$ is a **plane-parallel Monte Carlo** RT; ours is an analytic escape-probability formula, so the two cannot agree quantitatively — a quantitative comparison is only sensible once *all* Phase-3b features (incl. the velocity term) are in. Qualitatively (lya_caseA_huang11.py), both show a significant inner peak (ATES ~ 0.6 at 1.35 , Huang ~ 0.5 at $1.6 R_p$) and comparable magnitude ($0.05\text{--}0.6$). ATES has a dip at $\sim 1.8 R_p$ and rises gently outward where Huang gently declines; the outward rise traces to the $n2s$ recombination cascade in the Phase-3a $\text{H}(2s/2p)$ model ($n=2$ stays recombination-populated while ground-state H vanishes in the ionized wind), *not* $\bar{J}$, which after the two fixes below is physically sound (J_{int} peaks then declines; $J_{\text{star}} \approx \text{const}$ outward). $\text{H}(n=2)$ heating stays $\sim 1\%$ of total with $\text{Hdx} \gtrsim \text{Hpe}$ near the n_2 peak (cf. Fig. 10).

Two $\bar{J}$ bugs fixed during the trend check. (1) J_{star} used the planetary $\Delta\nu_D$ instead of the stellar width $\Delta\nu_*$, making $\bar{J}$ rise as the wind cooled; (2) J_{int} lacked the $(1-\beta)$ closure, so S diverged ($\propto 1/n_{1s}$) in the ionized wind.

Wind-velocity (Sobolev) escape — implemented, negligible here. $\beta_{\text{tot}} = 1 - (1 - \beta_{\text{static}})(1 - \beta_{\text{Sob}})$ with $\tau_S = C_{\text{sob}} n_{1s} / |dv/dr|$ is now in jlya_escape_prob (the outflow $v(r)$ threaded through excited_H_update). For WASP-121b it changes $\bar{J}$ by $< 1\%$: where it could help (thick inner wind) $\tau_S \sim \tau_{\text{static}} \sim 10^7$ so $\beta_{\text{Sob}} \sim 10^{-7}$, and the thin outer wind already has $\beta_{\text{static}} \approx 1$ — the velocity gradient and the $\text{Ly}\alpha$ opacity scale together ($\propto n_{1s}$), opening no new escape route. **Phase 3b is thus feature-complete.** *Deferred per the trend-first guidance:* the post-all-features quantitative comparison and the $n2s$ outward-rise revisit (a Phase-3a $2s$ effect, not the RT); boost, Δv_s , and the $\pi^{-1/4}$ prefactor are left un-tuned.

13. Phase 4: RLOF / tidal potential (Case B validated; Case D blocker = escape radius, transonic IC added, geometry-limited)

Phase 4 adds the Roche-lobe-overflow (RLOF) tidal physics that drives Huang’s enhanced mass loss — **Case B** $\dot{M} = 0.32$, **Case D** $= 1.03 M_p/\text{Gyr}$ — the regime that matches the observed transit depths (spherical Case A underestimates them). **ATES already has the 1D Roche potential** (grav_field.f90, the spherical_domain=.false. branch: planet + stellar-tidal + centrifugal), the L1-truncated domain, and the tidal momentum source (Source.f90) — Caldiroli’s ATES-v2 Roche mode. So the substellar tidal hydrodynamics is already in place.

13.1 Per-case base gravity (Huang Table 3)

The key to each case is the $1 \mu\text{bar}$ base radius R_0 . Huang’s Table 3 lists $\log g @ 1 \mu\text{bar} = GM_p/r_{1\mu\text{bar}}^2$ per case — exactly the R_0 -setting quantity ($R_0 = \sqrt{GM_p/g}$):

Case	tidal	$M_p (M_J)$	dT	XUV/Ly α /Fe	$\log g$	$R_0 (R_J)$
A	N	1.1824	0	1/1/1	2.84	2.058
B	Y	1.1824	0	1/1/1	2.70	2.418
D	Y	1.1204	350	0.74/0.35/4	2.62	2.581

R_0 grows A→B→D as the tidal force + the $dT+350$ lower-atmosphere heating puff the $1 \mu\text{bar}$ level outward (weaker base gravity), driving the RLOF.

13.2 Case B — validated

WASP-121b in Roche mode (tidal on) + the corrected base + metals + Ly α mode-2 gives $\dot{M} = 0.337 M_p/\text{Gyr} \approx$ Huang’s Case B (0.32), within $\sim 5\%$. With the base-radius fix both cases agree (Case A $0.077 \approx 1.5 \times$ Huang’s 0.052; Case B $0.337 \approx 1.05 \times$ Huang’s 0.32). ATES’s existing Roche potential thus captures the RLOF enhancement — no geometric-reconstruction code was needed for the Case B $\dot{M}$ gate.

13.3 Case D — real blocker found (escape radius), transonic IC added

Case D’s base ($R_0 = 2.581 R_J$, $\log g = 2.62$) sits at $\sim 76\%$ of the L1 distance: the L1-truncated Roche domain reaches only $r_{\text{max}} = 1.353 R_p$. Re-investigating the earlier “won’t launch” report uncovered the *actual* blocker, and it is *not* the initial condition.

Root cause — the escape radius lies outside the L1-truncated domain. `define_grid` sets the constant-momentum convergence window $[j_{\text{min}} : N]$ from the **escape radius** (Escape radius `[R_p]:`, default 1.50). It scans for the first cell with $r(j) \geq r_{\text{esc}}$; if *no* cell qualifies — which happens precisely when $r_{\text{esc}} > r_{\text{max}}$ — the DO loop falls through leaving $j = N + N_g + 1$, so $j_{\text{min}} > N$ and the window is **empty**. Fortran `maxval/minval` over a zero-size array return $\mp \text{HUGE}$, so $\text{mom}_{\text{max}} = -\text{HUGE}$, $\text{mom}_{\text{min}} = +\text{HUGE} \Rightarrow du = \text{Inf}$, and the time-derivative norm $dtu = -\text{HUGE}$. Since $dtu = -\text{HUGE} < dtu_{\text{th}}$, the loop reports **“steady state reached” at step 0** — a *false* convergence unrelated to the wind. The earlier bracket (“works $R_0 \leq 2.30$, fails $R_0 \geq 2.40$ ”) is exactly r_{max} crossing the 1.50 escape radius at $R_0 \approx 2.37$, confirming the escape radius, not the IC, as the discriminant. (The earlier “isothermal IC cannot launch the deep-RLOF wind” diagnosis was therefore wrong.)

Fixes. (1) A guard in `define_grid.f90`: if $j_{\text{min}} > N$, re-anchor the window to mid-domain ($r \geq 1 + \frac{1}{2}(r_{\text{max}} - 1)$) and print a loud WARNING — so this can never again masquerade as convergence. (2) For Case D, set Escape radius `[R_p]:` 1.20 (inside the 1.353 domain). With a valid window the diagnostics become finite and meaningful, and **Case D launches and runs** — from *both* the hydrostatic and the transonic IC. (The earlier $du = |(\text{mom}_{\text{max}} - \text{mom}_{\text{min}}) / \max(\text{mom}_{\text{min}}, 10^{-30})|$ floor at `ATES_main.f90:308` is retained as defense-in-depth against a genuine transient

zero-momentum cell, but it was *not* what unblocked Case D — with an empty window $\text{mom}_{\min} = +\text{HUGE}$, not zero, so the floor never engaged.)

Transonic-wind IC (implemented, Transonic IC: True). `set_IC.f90` gained an optional transonic isothermal-wind profile, solved from the ATES potential via the algebraic (Bernoulli) integral of the steady isothermal-wind equation,

$$\frac{1}{2}v^2 - c^2 \ln v - 2c^2 \ln r + \phi(r) = B, \quad c^2 = (1 + \delta p_{bc}) / \rho_{bc},$$

with B fixed by the sonic point ($v = c_s$ where $D\phi(r_c) = 2c^2/r_c$, located by bisection). At each radius the relation has a subsonic ($v < c_s$) and a supersonic ($v > c_s$) root; the wind takes the subsonic root for $r < r_c$ and the supersonic root for $r > r_c$ (1-D bisection on each monotone branch), and ρ then follows from mass conservation $\rho v r^2 = \dot{M}$ pinned to the base BC. Because that IC already satisfies $\rho v r^2 = \text{const}$, $du \approx 0$ at step 0 would itself trip the momentum-constant exit before the cold wind heats; the flag therefore **auto-enables force_start**. If no interior sonic point exists it **falls back to the hydrostatic IC**. Default **off** $\Rightarrow$ existing cases byte-identical (Case B re-verified at $\dot{M} = 0.337$).

Case D outcome — launches, but the 1-D L1-truncated domain undershoots. With the escape-radius fix the wind launches and relaxes, but it **never reaches a sonic point inside the domain**: the heated metal wind is **deeply subsonic everywhere** ($\text{Mach} \lesssim 0.05$ — rising to ~ 0.050 at $r \approx 1.25$, then *falling* to ~ 0.040 at L1). With a subsonic outer boundary the zero-gradient outflow BC is only marginally well-posed and $\rho v r^2$ stays non-constant (peaks $\sim 1.7\times$ the L1 value), so the run settles into a **breathing limit cycle**: after the $\sim 10^2$ heating transient damps, du keeps oscillating in $\sim 0.04\text{--}0.8$ (touching but never crossing the strict $du_{\text{th}} = 0.02$) for as long as it runs. The mass flux is correspondingly time-variable — the quasi-steady $\dot{M}$ oscillates over $\sim 0.3\text{--}0.6 M_p/\text{Gyr}$, well *below* Huang’s 1.03, the deficit set by the L1 truncation cutting off the still-accelerating subsonic wind before it reaches the sonic point. The transonic IC behaves identically to the hydrostatic IC here: both launch the same subsonic flow. (Warm-starting via `load_IC` reintroduces a base NaN — the documented grid re-read inconsistency — so it is not a workaround.)

Conclusion. Case D’s deep RLOF ($\dot{M} = 1.03$) is **not reachable in ATES’s 1-D L1-truncated Roche mode**: the wind is subsonic at L1, so the truncated domain structurally undershoots. Capturing it needs the 3-D geometric treatment (Phase 5) or a domain extended past L1 — *not* a different IC. The transonic IC and the escape-radius guard are correct, necessary infrastructure (they remove the false convergence and let the deep-RLOF wind launch and run with meaningful diagnostics) but are not sufficient on their own for Case D’s full mass-loss rate. The geometric reconstruction (substellar $\rightarrow$ terminator mapping, triaxial radii Table 2, the 4/9 mass-loss conversion) feeds Phase 5 (§14), and also supplies the geometry this subsonic L1-truncated wind needs.

14. Phase 5: velocity-broadened transmission spectrum (in progress)

Phase 5 is the headline deliverable — the NUV/optical **transit-depth spectrum** the whole upgrade exists to match (Huang Table 3 / Figs. 17–24). It extends the existing spherical post-processor `TPM.py` in two stages: **5a** adds the metal

resonance lines (spherical, gated on Case A); **5b** replaces the spherical geometry with the 3-D Roche-equipotential reconstruction + velocity broadening (gated on Case D). Pure Python post-processing — no Fortran change.

Line	Case A	Case D	Observed
Mg II $\lambda 2796$ (4 Å)	0.182	0.302	0.309 ± 0.036
Ca II K $\lambda 3934$	0.199	0.278	0.281 ± 0.009
H α $\lambda 6563$	0.201	0.185	0.186 ± 0.003
H β $\lambda 4861$	0.174	0.135	0.143 ± 0.005
Na D2 $\lambda 5890$	0.152	0.147	0.147 ± 0.002

These are the **effective transit radius** R_{eff}/R_* , not an absorption percent (the expected-results doc states the unit is R_p/R_*). Our code returns the line-center absorption fraction h , mapped to the same quantity via $R_{\text{eff}}/R_* = \sqrt{(R_p/R_*)^2 + h}$.

14.1 5a — metal resonance lines (Mg II, Ca II, Na I D), spherical — done

TPM.py gained Mg II $\lambda 2796$, Ca II K $\lambda 3934$, Na I D2 $\lambda 5890$ as an **isolated appended block** (the validated He 10830 / Ly α / H α / H β pipeline above it is untouched). These are resonance lines whose lower level is the ion ground state; at $\sim 10^4$ K the excited fine-structure levels are Boltzmann-negligible, so the lower-level density is the ion density itself ($n_{\text{lower}} \approx n_{\text{ion}}$). The ion densities live past the H/He block of Ion_species_adv.txt, in mion_fsp order (species_table.f90): numpy columns **17 = Mg II**, **23 = Ca II**, **25 = Na I**. The block reuses the existing spherical chord / Voigt (wofz) / trapezoid- τ / disk-average / instrument convolution machinery (resonance_depth()).

Case A result (line-center % / 4 Å-band %) and diagnosis. Mg II: 26.1 / 2.3 ($R_{\text{eff}} = 4.2 R_p$); Ca II K: 5.5 / 0.33 ($1.9 R_p$); H α : 3.9 ($1.6 R_p$); Na D2: 0.8 / 0.06 ($0.7 R_p$). Two real gaps: (1) **Mg II already reaches Huang's extent** ($R_{\text{eff}} = 4.2 R_p$ vs his clustered $\sim 3.6 R_p$) — the metal machinery is sound — but **Ca II / Na I / H α are too optically thin** ($R_{\text{eff}} = 0.7\text{--}1.9 R_p$), a line-specific ionization / abundance / $n = 2$ question, not geometry. (2) The lines are **too narrow**: the Case A wind reaches only ~ 29 km/s, so a 26%-deep core washes out to $\sim 2\%$ over a 4 Å (± 215 km/s) bin, whereas Huang's lines stay deep *across* the bin — they are velocity-broadened to hundreds of km/s by the fast Roche outflow + rotation, i.e. **Phase-5b physics**. (The existing H α tool shows the same shortfall, so it is systematic.) *Units — resolved*: Huang's table is R_{eff}/R_* (radius ratio; the expected-results doc header states the unit is R_p/R_*), not absorption percent. The Case A spherical run over-extends ($15 R_p$ domain $\Rightarrow$ Mg II reaches $4.2 R_p$, $R_{\text{eff}}/R_* = 0.51$ vs Huang 0.182); the proper test is the L1-confined Roche run in R_{eff}/R_* units (§5b).

14.2 5b — Roche-equipotential reconstruction + broadening — implemented

New module roche_recon.py. Assuming a uniform state on Roche-equipotential surfaces, the field at any (x, y, z) equals the 1-D substellar state at the equivalent radius r_{eff} where $\phi_{\text{sub}}(r_{\text{eff}}) = \phi_{3D}(x, y, z)$. The dimensionless Roche potential (R_p units) mirrors grav_field.f90:

$$\phi(x, y, z) = -\frac{1}{r} - \frac{q}{r_s} - \frac{1+q}{2a^3} [(x-x_{\text{cm}})^2 + y^2], \quad q = M_*/M_p, \quad a = \text{orbit}[R_p].$$

Tidal terms $\rightarrow 0$ recovers $\phi = -1/r$ (spherical), so the triaxial reconstruction reduces to the 5a spherical case for Case A. The module gives `roche_phi`, the L1 root of $d\phi_{\text{sub}}/dr = 0$, a monotone $\phi_{\text{sub}} \rightarrow r_{\text{eff}}$ inverse (ReconMap), the triaxial radii, and the Eggleton lobe radius. TPM.py gained a `geometry='triaxial'` switch (`triaxial_depth()`): the transit LOS runs along $+x$; the state at each 3-D point is the substellar state at r_{eff} ; and the **per-sector LOS velocity** $v_{\text{sub}}(r_{\text{eff}})x/r - \Omega y$ (wind + tidally-locked rotation, Huang Eq. 16) is integrated over **20 angular sectors**.

Demonstrated on the converged Case B (Roche) run. Geometry self-consistent: $L1 = 1.574 R_p$; $R_{\text{RL}}/L1 = 0.714 \approx 2/3$; tidally triaxial $R_{px} = 1.05 > R_{py} = 0.919 > R_{pz} = 0.893$ (substellar bulge, polar compression).

Result — the model reproduces Huang’s transit radii. Converting the Case B triaxial depths to R_{eff}/R_* (the table unit):

Line	Huang A	Huang D	Case B (this work)
Mg II	0.182	0.302	0.234
Ca II	0.199	0.278	0.222
Na D2	0.152	0.147	0.163
H α	0.201	0.185	0.239
H β	0.174	0.135	0.232

Case B is the *intermediate* model (between Case A and Case D), and the **metals (Mg II, Ca II, Na) land right between Huang’s Case A and Case D** — the lobe-confined reconstruction correctly reproduces his transit radii in magnitude and trend. The Balmer lines run $\sim 25\%$ high (H α 0.239 vs the expected ~ 0.19), consistent with the 5a finding that $n = 2/T$ is slightly over-predicted.

Correction. An earlier draft claimed the lobe-confined model “undershoots Huang because it omits the beyond-L1 escaping tail.” That was wrong twice over: (1) Huang’s reconstruction is *also* lobe-confined (equipotential mapping to the Roche-lobe boundary, §1.8) — neither model has a beyond-L1 tail, so it cannot be the difference; and (2) the apparent shortfall was a **units error** (our absorption *percent* vs Huang’s R_{eff}/R_*). In the correct units the two agree (table above). The small triaxial-vs-spherical reduction (terminator compression $R_{py} < R_{px}$) is minor; Huang’s area-preserving $R_{py}R_{pz} = R_p^2$ normalization would offset it.

14.3 5b — direct Case A comparison: hydro validated, metals over-extended

A clean spherical Case A at the *correct* 1 μbar base $R_0 = 2.058 R_J$ (Table 3 $\log g = 2.84$; the old `output_caseA` used $R_0 = 1.766$, the 4 mbar transit radius) was run to convergence.

- **Hydro / base validated:** $\dot{M} = 0.056 M_p/\text{Gyr} \approx$ Huang’s Case A 0.052 (the wrong base gave a different $\dot{M}$) — base and wind are right.
- **Metal transit radii — consistent once compared like-for-like** (Huang quotes Mg II in a 4 Å NUV bin; Ca II / Na are optical line-center): Mg II (4 Å) **0.220** vs Huang 0.182 ($\times 1.2$); Na D2 (LC) **0.175** vs 0.152 ($\times 1.15$); Ca II K (LC) **0.306** vs 0.199 ($\times 1.5$, high).

The “over-extension” was a comparison-definition artifact, not physics. An earlier draft reported $\text{Mg II } R_{\text{eff}}/R_* = 0.60$ ($3.3 \times$ Huang) — that compared the *line-center* R_{eff} against Huang’s $4 \text{ \AA}$ NUV-bin value. In the same $4 \text{ \AA}$ bin $\text{Mg II} = 0.22 \approx$ Huang’s 0.182; Na matches too; only Ca II is genuinely $\sim 1.5 \times$ high.

The metal ionization is correct and is NOT the lever. All channels are wired and physical (Verner+1996 σ , Badnell RR+DR, Voronov, Huang+2023 charge exchange; $a_\tau = 0$). The Mg II/III balance is a close competition between a weak photoionization ($\Gamma \approx 2 \times 10^{-3} \text{ s}^{-1}$, weak because $\sigma_{\text{MgII}} \approx 0.24 \text{ Mb}$ at threshold) and $\text{RR} \cdot n_e$; DR ($\sim 5 \times 10^5 \text{ K}$ resonances) and collisional ionization ($e^{-15 \text{ eV}/kT}$ at 10^4 K) are negligible. A test showed $\times 30$ **more Mg II photoionization moves R_{eff} only $0.57 \rightarrow 0.32$** — the resonance line is so optically thick that ionizing it harder cannot confine it, so the ionization is a dead end as a lever (and is not the cause once the bin is matched).

Status: the full Phase-5 machinery is implemented and runs end-to-end; the hydro/ $\dot{M}$ reproduces Huang (Case A 0.056, Case B 0.337); and the metal transit radii match Huang once compared in his own bins (Mg II $4 \text{ \AA}$ 0.22 vs 0.182, Na 0.175 vs 0.152). Remaining items are minor: **Ca II $\sim 1.5 \times$ high** (line-center; plausibly Ca II $\rightarrow$ Ca III or the Ca abundance), the slightly-high Balmer $n = 2$, the $R_{\text{py}} R_{\text{pz}} = R_p^2$ normalization (methodological), a terminating Case D run, and phase5_transmission.ipynb. Comparisons must use Huang’s per-line definition ($4 \text{ \AA}$ bin for the NUV metals, line-center for the optical lines) — the two differ by $\sim 3 \times$ for the optically-thick Mg II.

15. Post-process advection guard at a “breathing” base (2026-06-07)

Symptom. The spherical-vs-Roche example figure (user manual Fig. 4, WASP-121b) showed a sharp temperature spike at $r \approx 1.1 R_p$ in the *advection-corrected* (`_adv`) Roche profile, with a sawtooth band just below it. The converged equilibrium (eq) profile is smooth there, so the spike is a post-processing artifact, not a feature of the hydro solution.

Diagnosis (two compounding causes).

1. *A breathing, inflowing base (1-D limitation).* WASP-121b nearly fills its Roche lobe, so the 1-D wind is subsonic at L_1 and the dense lower atmosphere settles into a slowly recirculating state with small *negative* (inflow) velocities and a stagnation point ($v = 0$) near the wind base ($r \approx 1.1$). This is intrinsic to spherical symmetry, not a bug.
2. *A spurious hot root in the metal-cooled post-processor.* The advection post-processor re-solves the per-cell energy/ionization balance assuming an outflow (each cell upwinds from the next-inner cell). In the dense base the non-monotonic metal line-cooling curve gives the energy equation a second, spurious *hot* root that the Newton (MINPACK) solver can land on; the upwind coupling then cascades one bad cell outward, producing the sawtooth and the overshoot spike. The pre-existing $2 \times$ band guard around eq T did not catch the $\sim 1.4 \times$ spike.

Fix (option c: eq fallback where the flow is not a clean outflow). In `post_process_adv.f90`, wherever $v \leq 0$ (the breathing inflow base) the post-processor now *skips* the advection correction and keeps the converged eq ionization and temperature. Three guards were added, all gated on `pp_metal_on` so the metals-off mode stays byte-identical: the no-He and He H/He advection loops (set the H/He fractions to eq and `cycle` when $v(j) \leq 0$ or $v(j-1) \leq 0$) and the per-cell temperature solve (set $T_{\text{out}} = T_{\text{in}}$ and `cycle` when $v \leq 0$). Keeping the inner cells at eq removes the spurious

root *and* breaks the upwind cascade at its source, while the genuine `_adv` correction is retained in the outflow above the stagnation point.

Validation (PP-only re-run on the converged Case B state). Rebuilt and re-ran the post-processor (Do only PP on the converged IC):

- eq reproduced: $\max |\Delta T|/T = 2.6 \times 10^{-3}$, $\max |\Delta v| = 0.015 \text{ km s}^{-1}$; $\log_{10} \dot{M} = 13.38$ unchanged ($\approx 0.34 M_p \text{ Gyr}^{-1}$); Mg II reproduced to 1.6% in the line-forming region.
- `_adv` base oscillation eliminated: temperature sign changes in $[1.02, 1.10] R_p$ dropped from **36 to 0**; the spike is gone; the eq $\rightarrow$ `_adv` handoff at $v = 0$ is continuous (0.7% step) and the few-percent wind correction is preserved.
- Spherical Case A ($v > 0$ everywhere) is untouched — the guard never triggers, so its `_adv` is byte-equivalent.

Cause 1 (the breathing base) remains a 1-D limitation but is now represented there by the smooth eq solution; resolving the inflow itself would require a 2-D/3-D model.

Documentation updates (this session). The user manual (`ATES_user_manual.tex`) gained: §1.3 *The gravitational potential* (spherical, Roche substellar $\phi(r)$ and $d\phi/dr$, and the 3-D off-axis form, matching `grav_field.f90`); §3.5 *Spectral coverage for metal photoionization* (neutral-metal thresholds down to K I at 4.34 eV/2856 Å, and how the power-law sub-grid extends below 13.6 eV vs. a loaded SED that must itself reach those wavelengths); §8 *Atomic and molecular data* (a by-process source table — Asplund09 abundances, Verner96 photoionization, Voronov97 collisional ionization, Hui & Gnedin97 H/He + Badnell06 RR/adf48 DR metals + Shull & Van Steenberg82 Ca I + Huang23 Fe, Huang23 charge exchange, AIOLOS+CHIANTI cooling, Christie13/Draine11/Huang17 excited-H, NIST line data; and the explicit note that the code is purely atomic/ionic with *no* molecular data); and the §4 *breathing-base caveat* documenting the issue and fix above. The Fig. 4 example was regenerated from the corrected `_adv` output.

16. Code-size summary: ATES-metal vs. the original ATES (2026-06-07)

A line-by-line comparison of the Fortran source against the pristine upstream code (ATES/ATES-Code-main) quantifies the extent of the in-house extension. *Pure code lines* (SLOC) exclude blank lines and comments (the comment stripper is string-aware, so a `!` inside a quoted string is not treated as a comment); the line-by-line diff additionally normalizes whitespace and case, so reformatting and comment-only edits are *not* counted as changes.

	Fortran files	code lines (SLOC)
Original ATES (ATES-Code-main)	47	3 444
ATES-metal	57	7 258
net change	+10	+3 814 ($\approx 2.1\times$)

Line-by-line diff (comments + blanks excluded; whitespace/case normalized): **3 908** code lines added in ATES-metal (new modules + the new side of modified lines) and **94** removed from the original (deleted + the old side of modified lines), for **4 002** total changed lines. At the file level, of the 47 original files **21 were modified** and **26 are unchanged** (none were removed), and **10 new modules** were added:

New module	Purpose	SLOC
radiation/charge_exchange.f90	charge-exchange network (Huang Table 4)	311
radiation/excited_hydrogen.f90	H($n=2$) model + Balmer feedback	300
radiation/opacity_models.f90	opacity dispatcher (A/C/P/T)	180
nonlinear_system_solver/System_HeH_metals.f90	coupled H/He+metals ionization (+analytic Jacobian)	171
nonlinear_system_solver/newton_solver.f90	analytic-Jacobian Newton (+hybrd1 fallback)	124
time_step/energy_semi_implicit.f90	semi-implicit energy solver	104
files_IO/metals_input_read.f90	parse metals.inp	89
files_IO/opacity_input_read.f90	parse opacity.inp	85
init/species_table.f90	metal-species metadata table	70
radiation/lya_rt.f90	Ly α escape-probability RT	68
new-module total		1 502

So the 10 new modules contribute 1 502 of the 3 908 added code lines; the remaining $\sim 2\,406$ added (and 94 removed) lines are edits spread across the 21 modified files (chiefly the ionization-equilibrium, cooling, cross-section, I/O, gravity/IC, and energy modules that wire the metals, opacity dispatcher, Roche potential, transonic IC, excited-H/Ly α physics, and the Brent/Newton solver upgrades into the existing solver). In total ATES-metal roughly *doubles* the original code base ($3\,444 \rightarrow 7\,258$ SLOC, $\approx 2.1\times$) and touches just under half of its files (21 of 47 modified, plus 10 new).

17. He 2³S triplet + metals simultaneous solve, Ly α -flux guard, and convergence workflow (2026-06-08)

Motivation. Until now the He metastable triplet (He 2³S, “HeITR”) and the trace-metal ionization network could not be solved in the same run: both placed an extra unknown at $x(4)$ of the coupled ionization residual, and `input_read.f90` allocated the metal unknowns only if `(thereis_metals .and. .not. thereis_HeITR)`. Modelling HD189733b with the triplet *and* solar metals requires both at once.

Merged solver (factual). New module `nonlinear_system_solver/System_HeH_TR_metals.f90` (`ion_system_HeH_TR_metals`, 84 SLOC). The H/He/triplet rows ($x_1 = \text{HII}$, $x_2 = \text{HeII}$, $x_3 = \text{HeIII}$, $x_4 = \text{He } 2^3\text{S}$) are taken verbatim from `System_HeH_TR`; the metal rows are taken verbatim from `System_HeH_metals` and shifted to $x(5+2(e-1))$. The two blocks share one electron density $n_e = n_{\text{HII}} + n_{\text{HeII}} + 2n_{\text{HeIII}} + \sum_e (n_{X^+} + 2n_{X^{++}})$ (the neutral triplet is excluded), and charge exchange is added through `cx_add_to_fvec`. Solved with `hybrd1`. Supporting edits: `charge_exchange.f90` gained a settable base index `cx_metal_base` (default 4; the merged dispatch sets 5) so the metal charge-exchange rows land at the right offset; `input_read.f90` sets $N_{\text{eq}} = 4 + 2n_{\text{melem}}$ when both are active; `ionization_equilibrium.f90` uses a generalized metal base index (`mbase = 4`, or 5 with HeITR) in the initial guess / solution scatter and a both-on dispatch branch; the new source was added to the Makefile. The radiation grid, cross sections, photo-heating, IC setup and post-processing already handled both species; only the equilibrium *solver* needed merging.

Regression & physical check. With metals on / triplet off the run is byte-identical to the pre-merge binary over 8 048 steps (the new branch is not taken). With both on, the coupled solve is clean (no NaN) and was exercised end-to-end on WASP-121b with the full stack (triplet + metals + excited-H + $\text{Ly}\alpha$), reaching a flat converged solution (peak He 2^3S fraction $\sim 5 \times 10^{-6}$).

$\text{Ly}\alpha$ escape-probability input guard (factual). `input_read.f90`: if `Jlya escape-prob: True (jlya_mode=2)` but `Stellar Lya flux [erg/cm2/s]`: is missing or ≤ 0 , the run now stops with an explanatory message rather than silently dropping the stellar beam (in mode 2 $\bar{J}_{\text{Ly}\alpha} = \bar{J}_{\text{int}} + \bar{J}_*$ with $\bar{J}_* \propto F_{\text{Ly}\alpha,*}$, so $F_{\text{Ly}\alpha,*} = 0$ would make the halfwidth/boost settings silent no-ops).

Convergence threshold and automated two-stage reconstruction (factual).

- `du_th` is now a runtime variable (was a compile-time parameter) with default $1.0\text{e-}3$ — the original ATES-Code-main value. ATES-metal had carried $2.0\text{e-}2$, at which a run flagged “converged” can still show $\sim 2\%$ spatial spread in the supersonic mass flux $\rho v r^2$.
- A new runtime variable `du_th_plm` plus the input line `du_th [PLM,WENO3]: <du_plm> <du_final>` enables an *automatic* two-stage run: `ATES_main.f90` starts in PLM and switches `rec_method` to WENO3 once $du < du_{\text{plm}}$ (or PLM stalls), then converges at $du < du_{\text{final}}$. This automates the manual PLM $\rightarrow$ Load IC+WENO3 workflow recommended in the ATES README. If $du_{\text{plm}} \leq du_{\text{final}}$ (or the line is absent) the run is single-stage.
- CFL is now runtime and settable via the input line `CFL: <value>`.

Other source-term changes (kept, with caveats). `energy_semi_implicit.f90` now damps the implicit energy update with $\partial F / \partial T = 1 + c |dC/dT|$ (was $\max(0, dC/dT)$), adding damping on the falling cooling branch; it removed one oscillation in testing but did *not* by itself fix the convergence behaviour below. `Cool_coeff.f90` now interpolates the cooling tables with monotone PCHIP (C^1) rather than linear (C^0); this was implemented to test an interpolation-kink hypothesis (not supported, below) and is retained as a smoothness improvement.

Convergence — provisional observations (not conclusions). *The following is a research log, deliberately tentative; several intermediate “non-convergence” readings in earlier sessions were later traced to methodology, not physics.* Two artifacts are recorded so they are not repeated: an over-short step cap (weak-wind cases judged “floored” near 25k steps while du was still descending — HD209458b later reached $du < 2 \times 10^{-2}$ near 50k steps), and a run-script bug that killed the backgrounding subshell rather than the ATES.x child, so runs continued orphaned and the reported du was a premature snapshot (use `kill -x ATES.x`). Because of these, the earlier framing that “metal line cooling destabilizes the wind” should be treated as *not established*. With the two-stage strict-`du_th` pipeline we currently see, across planets:

- **WASP-121b** (inflated, strongly irradiated): converges quickly and flat ($\sim 0.16\%$ supersonic mass-flux spread, vs $\sim 2\%$ for single-stage PLM). This case looks solid.

- **HD209458b**: reaches a small $du \sim 0.02\text{--}0.025$ that then oscillates rather than dropping to 10^{-3} . *Tentatively* usable as a quasi-steady state: a He $2^3\text{S}+7$ -metal snapshot from the two-stage pipeline gives a smooth transonic wind with $T_{\text{max}} \approx 6.8 \times 10^3 \text{ K}$ and $\log_{10} \dot{M} \approx 10.1$ (median over the supersonic region), consistent with the literature $\sim 10^{10} \text{ g s}^{-1}$; see `heitr_metals_dev/HD209458b_quasisteady.ipynb`.
- **HD189733b** (strongly bound, $\beta_0 \approx 192$): du oscillates around ~ 2 in a sustained limit cycle and does not reach the threshold — the problematic case.

A plausible (but unproven) reading is that the difficulty tracks how weakly driven / strongly bound the wind is, rather than the metal content per se.

Base-breathing hypothesis and the fixes that did *not* work. The HD189733b oscillation resembles the known base-breathing behaviour, and the lower boundary (`Apply_BC.f90`, `BC_component_constrho`) hard-pins the ghost density and pressure to cold reservoir values with a one-way velocity valve $\max(v_1, 0)$ — a configuration that can reflect acoustic waves. Three variants were tried and reverted:

Test	Outcome
Lower CFL ($0.6 \rightarrow 0.2$)	du floor barely moved ($\sim 2.8 \rightarrow \sim 1.9$)
Zero-gradient base velocity (drop the valve)	du still ~ 2.3
Riemann-invariant base velocity $v = v_1 + \frac{2(c_p - c_l)}{\gamma - 1}$	did not help HD189733b <i>and</i> broke WASP-121b

So the base-breathing reading is *consistent with* the observations but is **not demonstrated**, and these simple BC variants are not solutions. Candidate routes if pursued (both substantial, uncertain payoff): a characteristic / non-reflecting (NSCBC-style) base BC; a steady-state Newton / BVP solver (which would not orbit a time-marching limit cycle if a steady solution exists); or accepting HD189733b as a marginal, time-averaged quasi-steady case. A separate, cheaper idea for *speed* (not the oscillation) is local per-cell time-stepping, since the global dt is currently set by the smallest base cell (`eval_dt.f90`). Full working notes: `docs/heitr_metals_and_convergence_notes.md`; solver-option discussion: `docs/numerical_methods.md`.

Code-size note. The merge adds one Fortran module (`System_HeH_TR_metals.f90`, 84 SLOC), so the 2026-06-07 file tally of 57 becomes 58; the remaining changes are small edits to `charge_exchange.f90`, `input_read.f90`, `ionization_equilibrium.f90`, `parameters.f90`, `ATES_main.f90`, `energy_semi_implicit.f90` and `Cool_coeff.f90`.

18. Metals in the bulk-mass/charge/particle budget (`eos_metals`, 2026-06-13)

Motivation. Until this update the trace metals were strictly *passive* species: they contributed to heating/cooling and were solved in the coupled ionization equilibrium, but they were excluded from the mass density, the base density ρ_{bc} , and the electron/particle densities used everywhere *outside* the ionization solver (EOS temperature, opacities, cooling calls, energy update). The only self-consistent place was the charge balance *inside* the MINPACK residuals

(System_HeH_metals.f90). At solar abundances the missing metal mass is $\sim 1.3\%$ of the H/He mass — small but systematic, and inconsistent with solving the metal ionization to high precision. Recorded in TO_BE_DONE.md on 2026-06-12; implemented 2026-06-13.

Runtime interface. A new key `eos_metals 0|1` in `metals.inp` (documented in `inputdata/metals.inp.example`). `1` (the new default) includes the metals in the bulk-gas budget; `0` restores the legacy trace approximation exactly. Metals-off runs (no `metals.inp`) are identical either way.

What changed (factual).

- **Composition constants** (`input_read.f90`): the mass per hydrogen nucleus is now $\mu_H \equiv \text{mass_per_H} = 1 + 4Y + \sum_X a_X A_X$ (with $Y = \text{HeH}$ and a_X, A_X the metal abundance and atomic weight), and the base density becomes $\rho_{bc} = \mu_H / (1 + Y)$ in units of $m_H n_0$. Atomic weights live in `species_table.f90:melem_A`. The normalization n_0 keeps its H/He-nuclei meaning, so $n_H = n_0 / (1 + Y)$ is unchanged. (Correction to the original TO_BE_DONE entry: v_0, p_0, b_0 are m_H -based normalizations and never contained a mean molecular weight — they were never affected.)
- **Mass density:** `calc_rho` takes an optional `nm` argument and adds $\sum A_X n_{X,i}$ over all stages; passed from `ionization_equilibrium`.
- **Electron density:** `calc_ne` takes an optional `nm` and adds $\sum q_i n_{X,i}$; passed at every site where `nm` is available — composition (main-loop EOS), `ionization_equilibrium` (opacity/cooling), `util_ion_eq` (eval_cool + cooling dump), `energy_semi_implicit`, `excited_hydrogen`, `post_process_adv`. (The scalar `T_equation` already carried metal electrons via `pp_nm_cell`.)
- **Particle density:** `calc_ntot` likewise adds the metal nuclei, so the EOS $T = p / (n_{\text{tot}} + n_e)$ sees them.
- **Ghost pressure / dp_bc:** the lower-BC ghost pressure is now `ntot_bc + dp_bc` with $n_{\text{tot},bc} = (1 + Y + \sum_X a_X) / (1 + Y)$ — so $T = T_0$ holds *exactly* at the base — and `dp_bc` includes the metal electrons $\sum q_i n_{X,i} / n_0$ at the first cell. `set_IC/load_IC` use `mass_per_H/ntot_bc` consistently. Full formulas: `ATES_BC_and_IC.tex`.

Validation (WASP-121b regression matrix).

- `eos_metals 0` is **byte-identical** to the pre-change goldens for both regression cases (`wasp_full 7296` steps, `wasp_he23off 7300`) — the legacy path is untouched.
- `eos_metals 1` (new default) converges cleanly (7145/7142 steps, $\sim 2\%$ fewer); profile shifts are at the expected $\sim 1\%$ mass scale: median $|\Delta n| \sim 2.4\%$, $|\Delta p| \sim 2.3\%$, $|\Delta T| \sim 0.9\%$ (max $|\Delta T| = 132$ K in the breathing-base region $r \sim 1.09$). The *number* flux nvr^2 drops 1.1% ($\log_{10} \dot{M}$: $13.31 \rightarrow 13.30$), which offsets the $+1.2\%$ mass per particle — the *mass*-loss rate is essentially unchanged.
- Goldens re-snapshotted under the new default; a repeat run reproduces them byte-identically (single-thread determinism intact).

Still open (small). The `use_2lev_cool` legacy cooling branch remains AIOLOS-hard-coded (default off), and the per-cell scalar `T_equation` uses `pp_nm_cell` only in the post-process path (the time-marching per-cell solve receives n_e via the now-metal-aware `energy_semi_implicit`).

19. Automatic initial-condition selection (IC mode: auto, 2026-06-13)

Motivation. The code now carries three IC families — cold hydrostatic (default), transonic isothermal wind (added for the Phase-4 deep-RLOF Case D, where the cold IC false-converges), and the hot-Parker warm seed (benchmarked: no speedup) — and the choice was manual. Kubyshkina+2018 build a per-planet IC automatically; the design study is `docs/auto_ic_design.md` and the literature comparison `docs/code_comparison.tex`.

The selector (one exact probe, no tunable threshold). `select_IC_auto` (`set_IC.f90`) evaluates the *cold* base sound speed in code units, $c_{\text{cold}}^2 = (n_{\text{tot,bc}} + \Delta p_{\text{bc}}) / \rho_{\text{bc}}$, and calls the existing `find_sonic` bisection for a critical point $\phi'(r_c) = 2c^2/r_c$ in the *actual* potential (Roche or spherical). If an interior sonic point exists, the cold isothermal atmosphere already wants to blow a transonic wind, so the transonic IC is selected; otherwise the cold hydrostatic IC is kept. One criterion catches both regimes that need the transonic start: deep RLOF (through the Roche/L1 topology) and low-gravity boil-off ($r_c \simeq b_0/(2c_0^2)$ in a point-mass potential). The Jeans parameter b_0 (Kubyshkina’s Λ) is logged as a diagnostic but is *not* a decision variable; the hot-Parker seed is never auto-selected.

Runtime interface. New input line `IC mode: <cold|transonic|hot_parker|auto>`. The named modes simply set the legacy flags; `auto` defers the decision to `set_IC`. The legacy keys (`Transonic IC:`, `Hot Parker IC:`) take precedence over `auto`, and the setup report (`ATES.out`) records the family actually in effect, with a “(chosen automatically: IC mode = auto)” tag. Touched files: `parameters.f90` (`ic_mode`), `input_read.f90` (key parsing), `set_IC.f90` (`select_IC_auto`), `write_setup_report.f90`.

Validation gates (all run 2026-06-13).

Gate	Result
A-1 no IC mode key	byte-identical regression (defaults untouched)
A-2 IC mode: cold	byte-identical to A-1
A-3 spherical tutorial, auto	$b_0 = 83.2$, no interior $r_c \rightarrow$ cold; IC_dump byte-identical
B-1 Case D (deep RLOF), auto	$r_c = 1.297$ (just inside $L1 = 1.353$) $\rightarrow$ transonic, same as the manual key
B-2 boil-off planet ($0.0189M_J$, $0.446R_J$, 1100 K)	$b_0 = 8.5$, $r_c = 5.2$ (as predicted) $\rightarrow$ transonic; wind launches (initial residual 0.19 vs 14 cold)
B-3 wasp_full (Roche), auto	$\rightarrow$ transonic; <i>du</i> -stop in 5876 vs 7145 steps (−18%)

Honest speed verdict. The −18% step count is *Roche/deep-RLOF-specific* (the steady wind there is close to the cold isothermal transonic solution). For the mild spherical boil-off case B-2 the cold control *also* launched and converged first ($\log_{10} \dot{M} = 11.20$), while the auto run relaxed monotonically toward the same flux from $\sim 2\times$ above — the cold- c^2 Parker profile *overestimates* the flux of a weakly heated wind. So auto is not harmful (same attractor) but not faster there.

Key byproduct (a trap for all $\dot{M}$ work). While adjudicating gate B-3 it was found that the $du < 10^{-3}$ stopping criterion is **path-dependent at the $\sim 5\%$ level**: the cold-start and transonic-start `wasp_full` runs both “converge” by du but sit 4.8% apart in mass flux, each only 1–2% flux-flat. Newton-finishing *both* states (Load IC + Solver: Newton) collapses them onto the *same* fixed point (flux ratio 0.9996, flatness 0.005%). Practical rule, now standing: **for quantitative $\dot{M}$ always finish with Solver: Newton**; never compare bare du -stops at the percent level.

Deferred. A Phase-C retry ladder (escalate cold $\rightarrow$ warm seed on NaN/stall) is designed in `auto_ic_design.md` §6 but *not* implemented — the B-2 result suggests failure-triggered escalation is the right shape if a real case ever needs it. IC formulas and the selection logic are also documented in `ATES_BC_and_IC.tex` §5 and the user manual.